



**De Montfort University**

**Final Year Project Dissertation**

**FinSentinel: A Governance-Aware Human–AI Collaborative Framework for Explainable  
Financial Fraud Detection**

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Academic Year: **2025–2026**

Date of Submission: **22 May 2026**

Word Count: 12,560

# Declaration

I declare that this dissertation is entirely my own work and that all sources have been appropriately acknowledged and referenced. I confirm that this work has not previously been submitted for any other degree or academic award.

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# Acknowledgements

I would like to express my sincere gratitude to my supervisor, **Dr David Elizondo**, for his continuous guidance, encouragement, and valuable feedback throughout the development of this project. His expertise and support significantly contributed to the successful completion of this research.

I would also like to extend my appreciation to my **second marker, Lipika Deka** for their time, constructive feedback, and evaluation of this work. Their observations and recommendations contributed towards strengthening the quality of the project.

I would further like to thank **De Montfort University**, the lecturers and academic staff for providing the knowledge, resources, and learning environment that supported my academic journey and the completion of this Final Year Project.

I am grateful to my family and friends for their patience, motivation, and continuous support throughout the duration of this project.

Finally, I would like to acknowledge the technologies, research resources, and open-source tools that supported the implementation, experimentation, and evaluation activities conducted within this study.

# Abstract

Financial institutions increasingly employ machine learning techniques to strengthen fraud detection capabilities; however, many existing approaches primarily emphasise predictive performance while providing limited transparency, explainability, and operational governance. Black-box behaviour, evolving fraud patterns, and the absence of human oversight can reduce trust and negatively affect practical deployment within financial environments. Therefore, there is a need for integrated systems capable of combining predictive intelligence with explainability, governance mechanisms, and continuous behavioural monitoring.

This project proposes **FinSentinel**, a governance-aware Human–AI collaborative fraud detection framework designed to improve transparency and operational reliability. The framework integrates machine learning-based fraud prediction, SHAP explainability mechanisms, threshold-based governance routing, behavioural drift intelligence, probability calibration, and an interactive dashboard environment. The Credit Card Fraud Detection dataset containing 284,807 transaction records was utilised for implementation and evaluation. Multiple machine learning models were comparatively evaluated, and Random Forest was selected as the primary operational model because of its balance between predictive performance and interpretability.

Experimental evaluation demonstrated strong predictive capability, with the selected model achieving a ROC–AUC score of **0.977**, PR–AUC of **0.814**, and F1-score of **0.814**. Explainability outputs successfully identified influential variables affecting prediction behaviour, while governance experimentation demonstrated trade-offs between fraud capture effectiveness and analyst workload. Behavioural drift monitoring further provided contextual intelligence regarding changing transaction patterns.

The primary contribution of this work involves the development of an integrated Human–AI collaborative framework that extends beyond conventional prediction-focused systems by incorporating explainability, governance intelligence, behavioural monitoring, and operational decision support within a unified environment. The proposed framework demonstrates how predictive intelligence and human oversight can operate together to improve transparency, trust, and long-term operational effectiveness.

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## Abbreviations

| Abbreviation | Meaning                             |
|--------------|-------------------------------------|
| AI           | Artificial Intelligence             |
| ML           | Machine Learning                    |
| XAI          | Explainable Artificial Intelligence |
| SHAP         | SHapley Additive Explanations       |
| RF           | Random Forest                       |
| MLP          | Multi-Layer Perceptron              |
| PCA          | Principal Component Analysis        |
| KPI          | Key Performance Indicator           |
| ROC          | Receiver Operating Characteristic   |
| AUC          | Area Under Curve                    |
| PR-AUC       | Precision–Recall Area Under Curve   |
| HITL         | Human-in-the-Loop                   |

# Introduction

## 1.1 Background

The rapid growth of digital technologies has transformed the financial sector by increasing the volume, speed, and accessibility of electronic transactions. Online banking systems, digital payment platforms, e-commerce services, and mobile financial applications have significantly improved transaction convenience for both individuals and organisations. However, the increasing dependence on digital financial infrastructures has simultaneously introduced new opportunities for cybercriminal activities and increasingly sophisticated fraudulent behaviour. Financial institutions continue to experience substantial operational and economic challenges associated with fraud-related activities, which may lead to financial losses, reputational damage, and reductions in customer trust (Ngai et al., 2011).

Traditional fraud detection approaches primarily relied on manually engineered rules and static verification procedures. These systems commonly operated using predefined conditions such as transaction amount thresholds, geographical restrictions, and known suspicious behavioural patterns. One advantage of rule-based systems involved their high degree of interpretability because decision-making processes could be traced directly to explicitly defined rules. However, such systems frequently demonstrated limited adaptability because they relied heavily on historical assumptions and manually constructed conditions. Fraudsters continuously modify behavioural patterns to bypass defensive mechanisms, reducing the long-term effectiveness of static rule-based approaches (West and Bhattacharya, 2016).

The emergence of machine learning techniques introduced more sophisticated approaches capable of identifying hidden patterns and complex relationships within transaction datasets. Machine learning algorithms including Logistic Regression, Random Forest, Extreme Gradient Boosting (XGBoost), and Neural Networks have demonstrated improved fraud detection capability compared with traditional systems. Unlike manually defined rules, machine learning

systems can automatically learn behavioural characteristics from historical observations and improve predictive performance through iterative learning processes (Dal Pozzolo et al., 2015).

Despite these improvements, high predictive performance alone does not necessarily guarantee practical deployment effectiveness. Many advanced machine learning systems operate as black-box models in which internal decision-making behaviour remains difficult to understand.

Although complex models may achieve strong classification performance, users often cannot determine why a transaction was classified as fraudulent or legitimate. Limited interpretability creates challenges within financial environments because automated decisions frequently require transparency and accountability. Such concerns become increasingly important in situations where incorrect classifications may negatively affect customers or organisational operations (Adadi and Berrada, 2018).

To address these limitations, Explainable Artificial Intelligence (XAI) techniques have emerged to improve transparency and understanding of machine learning predictions. Explainability approaches attempt to identify the contribution of variables toward generated outcomes and provide interpretable representations of model behaviour. SHAP (SHapley Additive Explanations) has become a widely adopted explainability technique because it provides mathematically grounded explanations capable of supporting both local and global interpretation of prediction outcomes (Lundberg and Lee, 2017).

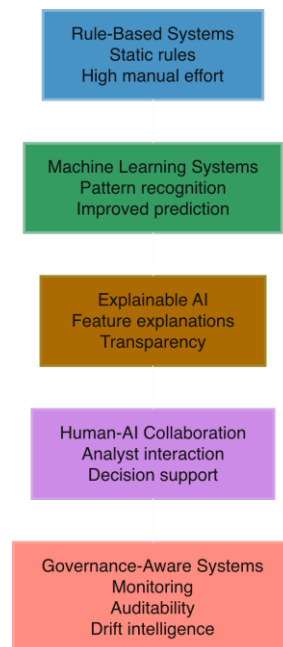
Recent research has additionally shifted attention from fully autonomous systems toward Human–AI collaborative environments. Human–AI collaboration combines machine intelligence with human judgement by introducing selective intervention mechanisms in situations involving prediction uncertainty. Rather than replacing human expertise completely, collaborative approaches attempt to utilise machine learning as a decision-support capability where human oversight remains integrated within operational workflows. Human involvement may improve accountability, reduce automation risk, and support more reliable decision-making processes (Amershi et al., 2019).

Furthermore, machine learning systems frequently assume that future observations follow patterns similar to historical training data. In practice, however, customer behaviour and fraud strategies continuously evolve over time. Such behavioural changes may gradually reduce model effectiveness and introduce concept drift conditions where predictive systems no longer accurately represent operational environments. Continuous monitoring mechanisms can therefore provide early identification of changing behavioural patterns and improve long-term operational reliability (Gama et al., 2014).

Consequently, modern fraud detection systems increasingly require more than predictive intelligence alone. Practical deployment environments demand explainability, governance

mechanisms, human oversight, and behavioural monitoring capabilities that can improve transparency, trust, and operational sustainability.

Figure 1.1: Evolution of Fraud Detection Approaches



*Figure 1.1 illustrates the progression of fraud detection approaches from traditional rule-based systems toward explainable and governance-aware Human-AI environments.*

## 1.2 Problem Statement

Financial institutions increasingly rely on machine learning techniques to strengthen fraud detection capabilities and improve predictive accuracy. Although many machine learning models demonstrate strong performance in identifying fraudulent transactions, existing approaches frequently prioritise predictive outcomes while providing limited support for transparency, interpretability, and operational governance. High-performing systems may therefore produce accurate classifications while simultaneously functioning as black-box environments where decision-making processes remain difficult to understand (Adadi and Berrada, 2018).

Limited interpretability introduces practical challenges within financial environments because analysts and stakeholders may hesitate to trust recommendations that cannot be adequately justified. Incorrect predictions may also result in financial loss, operational inefficiencies, or reduced customer satisfaction. Furthermore, fully automated environments may introduce additional risks where uncertain predictions are processed without sufficient human intervention.

Existing approaches also commonly assume stable operational environments despite fraud behaviour continuously evolving over time. Behavioural changes may gradually reduce model effectiveness and introduce concept drift conditions that negatively affect prediction reliability (Gama et al., 2014).

Therefore, a need exists for an integrated framework capable of combining predictive intelligence with explainability, governance mechanisms, human oversight, and behavioural monitoring capabilities in order to improve transparency, operational trust, and long-term reliability.

## 1.3 Aim

To develop a governance-aware explainable Human–AI collaborative fraud detection framework integrating predictive intelligence, operational routing, human oversight, and behavioural drift intelligence.



## 1.4 Objectives

Table 1.1: Project Objectives

| Objective | Description                                                      |
|-----------|------------------------------------------------------------------|
| O1        | Develop fraud detection models using machine learning techniques |
| O2        | Compare predictive performance across multiple models            |
| O3        | Integrate explainability mechanisms into model predictions       |
| O4        | Develop governance routing logic for operational decisions       |
| O5        | Implement behavioural drift monitoring mechanisms                |
| O6        | Develop Human–AI collaborative workflows                         |
| O7        | Build an interactive operational dashboard                       |
| O8        | Evaluate operational effectiveness using performance indicators  |

*Table 1.1 summarises the objectives established to achieve the overall project aim.*

## 1.5 Research Questions

Table 1.2: Research Questions

| ID  | Research Question                                                             |
|-----|-------------------------------------------------------------------------------|
| RQ1 | Can governance-aware AI improve operational trust in fraud detection systems? |
| RQ2 | Can selective escalation reduce operational risk?                             |
| RQ3 | Can explainability improve decision transparency?                             |
| RQ4 | Can behavioural drift intelligence improve long-term operational reliability? |

*Table 1.2 presents the research questions investigated within the project.*

## 1.6 Research Hypotheses

Table 1.3: Research Hypotheses

| Hypothesis | Description                                                  |
|------------|--------------------------------------------------------------|
| H1         | Governance-aware systems improve operational trust           |
| H2         | Selective escalation reduces operational risk                |
| H3         | Explainability improves model interpretability               |
| H4         | Behavioural drift intelligence improves long-term robustness |

*Table 1.3 presents the hypotheses developed to support investigation of the research objectives.*

## 1.7 Scope

Table 1.4: Project Scope

| Included                             | Excluded                        |
|--------------------------------------|---------------------------------|
| Fraud prediction using ML models     | Real-time banking deployment    |
| SHAP explainability integration      | Production cloud deployment     |
| Human–AI collaborative routing       | Live analyst feedback systems   |
| Governance threshold experimentation | Multi-bank datasets             |
| Behavioural drift monitoring         | Enterprise-scale infrastructure |
| Dashboard implementation             | Full cybersecurity coverage     |

*Table 1.4 summarises the project scope by identifying included and excluded areas.*

## 1.8 Personal Development Objectives

The project supported development in machine learning, explainable AI, governance design, and full-stack implementation. The development process strengthened skills in data preprocessing, model development, dashboard integration, performance evaluation, and research analysis. Exposure to Human–AI collaboration and governance principles additionally improved understanding of practical challenges associated with AI deployment.

## 1.9 Report Structure

This dissertation is organised into eight chapters covering the research problem, framework development, evaluation, and reflection activities. The report progresses from theoretical foundations toward implementation and experimental analysis before discussing findings, limitations, and future improvements.

# Literature review

## 2.1 Evolution of Fraud Detection Approaches

Fraud detection mechanisms have evolved significantly due to the increasing complexity of financial transactions and the growing sophistication of fraudulent behaviour. Early fraud detection systems relied primarily on rule-based techniques where manually defined conditions were used to identify suspicious activities. These systems commonly incorporated static rules such as transaction amount limits, geographic restrictions, or predefined behavioural patterns. Rule-based approaches offered advantages including high interpretability and straightforward implementation because decision logic remained explicitly visible to analysts. However, such approaches frequently demonstrated limited adaptability because new fraud strategies required continuous manual updates and rule redesign (Ngai et al., 2011).

As transaction volumes increased and fraudulent behaviours became increasingly complex, machine learning techniques emerged as more effective alternatives. Machine learning systems introduced the ability to automatically identify hidden relationships within large datasets and detect complex behavioural patterns without requiring manually specified rules. Algorithms including Logistic Regression, Decision Trees, Random Forests, Support Vector Machines, and Multi-Layer Perceptron (MLP) models demonstrated improved predictive capabilities and stronger adaptability to evolving environments (West and Bhattacharya, 2016).

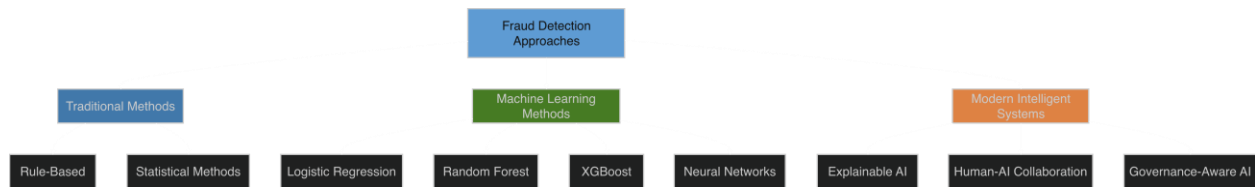
Despite performance improvements, increasing model complexity introduced additional challenges related to transparency and interpretability. Complex algorithms frequently operate as black-box systems where users may struggle to understand the rationale behind generated predictions. Such limitations become particularly significant within high-risk environments such as fraud detection, where operational decisions require transparency and accountability (Adadi and Berrada, 2018).

Recent developments therefore indicate a shift toward explainable and collaborative approaches. Explainable Artificial Intelligence (XAI) techniques attempt to improve understanding of model behaviour by generating interpretable representations of decision-making processes. Simultaneously, Human–AI collaborative systems integrate human expertise into prediction workflows to improve accountability and support operational trust. More recently, governance-focused systems incorporating monitoring and behavioural drift intelligence have emerged to maintain long-term system reliability within changing environments (Amershi et al., 2019).

The progression of fraud detection approaches demonstrates a shift from static rule-based systems towards adaptive and explainable frameworks. However, existing systems frequently

prioritise predictive performance while treating explainability, governance, and behavioural monitoring as independent components. This limitation motivates the development of integrated operational frameworks capable of supporting both predictive intelligence and trustworthy decision-making.

Figure 2.1: Fraud Detection Taxonomy



*Figure 2.1 categorises fraud detection approaches from traditional rule-based methods toward modern intelligent and governance-aware systems.*

## 2.2 Comparative Analysis of Machine Learning Models

Machine learning techniques have become widely used within fraud detection systems because of their capability to identify hidden relationships and complex behavioural patterns within large datasets. Various algorithms demonstrate different strengths and limitations with respect to predictive accuracy, computational efficiency, interpretability, and operational suitability. Since fraud detection environments often require both strong predictive performance and explainability, selecting an appropriate model involves evaluating multiple characteristics rather than relying solely on classification accuracy.

**Logistic Regression** is among the most widely used statistical classification approaches and has frequently been applied within fraud detection studies because of its simplicity and high interpretability. The model estimates the probability of class membership through linear relationships between input variables and predicted outcomes. Logistic Regression generally performs effectively when relationships between variables remain relatively straightforward and computational efficiency is important. However, fraud datasets frequently contain complex non-linear relationships that may reduce the predictive capability of linear models (Dal Pozzolo et al., 2015).

**Random Forest** extends decision-tree approaches by combining multiple trees into an ensemble structure. Ensemble learning improves prediction robustness and reduces overfitting problems frequently associated with individual decision trees. Random Forest models can capture non-linear relationships and interactions among variables while maintaining relatively good interpretability through feature importance analysis (Biau and Scornet, 2016). However, large ensembles may increase computational complexity and reduce transparency compared with simpler statistical methods (Breiman, 2001).

**Extreme Gradient Boosting (XGBoost)** has demonstrated strong predictive performance across many classification problems because it combines boosting techniques with regularisation mechanisms that improve learning efficiency and reduce overfitting. XGBoost frequently achieves high classification accuracy in fraud detection environments due to its ability to model complex interactions among variables. Nevertheless, increasing model complexity may reduce interpretability and create challenges in operational environments requiring transparent decision-making (Chen and Guestrin, 2016).

Multi-Layer Perceptron (MLP) models provide flexible non-linear learning capability through interconnected hidden layers capable of identifying complex relationships among variables. MLP architectures have demonstrated strong predictive performance across classification problems because they can learn interactions extending beyond linear assumptions. However, MLP models frequently require additional hyperparameter optimisation and may reduce interpretability because internal decision processes become less transparent compared with simpler approaches

such as Logistic Regression or Random Forest. Within operational fraud environments requiring explainability and governance support, reduced transparency may create practical deployment challenges (LeCun et al., 2015).

Although predictive accuracy remains important, operational deployment considerations require additional evaluation criteria including interpretability, transparency, computational efficiency, and governance suitability. Therefore, selecting models for fraud detection systems should involve balancing predictive capability with practical deployment requirements rather than focusing exclusively on performance metrics.

Table 2.1: Comparative Analysis of Machine Learning Models for Fraud Detection

| Model                        | Strengths                                 | Weaknesses                                   | Interpretability | Operational Suitability |
|------------------------------|-------------------------------------------|----------------------------------------------|------------------|-------------------------|
| Logistic Regression          | Fast, simple implementation               | Limited non-linear capability                | High             | Moderate                |
| Random Forest                | Robust, handles complex patterns          | Larger computational cost                    | Moderate         | High                    |
| XGBoost                      | Strong predictive performance             | Reduced transparency                         | Moderate-Low     | High                    |
| Multi Layer Perceptron (MLP) | Captures complex non-linear relationships | Requires tuning and reduced interpretability | Low              | Moderate                |

*Table 2.1 summarises the comparative characteristics of selected machine learning approaches used in fraud detection environments.*



## 2.3 Explainable Artificial Intelligence (XAI)

As machine learning systems become increasingly sophisticated, concerns regarding transparency and interpretability have become more significant, particularly in high-risk domains such as financial fraud detection. Many advanced algorithms including ensemble models and Neural Networks frequently operate as black-box systems where users may observe prediction outcomes without understanding the underlying decision-making process. Although such models often demonstrate high predictive performance, limited interpretability may reduce user trust and create challenges for operational adoption (Adadi and Berrada, 2018).

Explainable Artificial Intelligence (XAI) refers to techniques developed to improve understanding of model behaviour by providing interpretable representations of prediction processes. Explainability mechanisms attempt to identify how input variables influence generated outcomes and support users in understanding why specific decisions were produced. Within financial environments, explainability has become increasingly important because prediction outcomes may influence sensitive operational decisions involving transaction approval, investigation processes, or customer interactions.

Several explainability approaches have been proposed within machine learning literature. One commonly used approach is **Local Interpretable Model-Agnostic Explanations (LIME)**, which generates local explanations by approximating complex models using simpler interpretable models. LIME provides useful insight into individual predictions; however, explanation consistency may vary because explanations depend on local sampling procedures (Ribeiro et al., 2016).

An alternative approach involves **SHapley Additive exPlanations (SHAP)**, which applies principles derived from cooperative game theory to determine the contribution of individual variables toward prediction outcomes. SHAP provides both local and global interpretability by assigning importance values to features according to their relative influence on predictions. Unlike some alternative methods, SHAP offers theoretical consistency and additive properties that improve explanation reliability and support more comprehensive model interpretation (Lundberg and Lee, 2017).

Within fraud detection systems, explainability may provide several operational benefits. Analysts can understand which features contribute most significantly toward suspicious transaction identification, enabling greater confidence in automated recommendations. Explainability can additionally support model validation by identifying unexpected behaviour and reducing the likelihood of incorrect decisions remaining unnoticed.

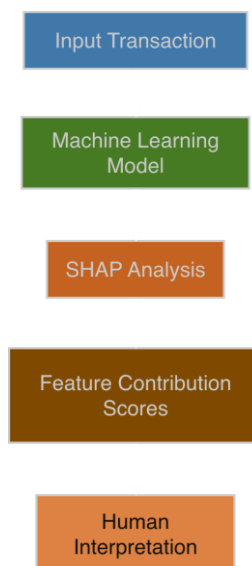
For the proposed FinSentinel framework, SHAP was selected as the primary explainability mechanism because it provides both transaction-level and model-level interpretation capabilities.

Compared with LIME, SHAP offers greater consistency and supports a broader understanding of prediction behaviour across operational environments. This capability aligns with the project objective of improving transparency and supporting governance-aware decision-making rather than relying exclusively on predictive accuracy.

Consequently, explainable mechanisms represent an important component within modern AI systems because they improve transparency, strengthen trust, and support more reliable Human–AI collaboration.

Although both LIME and SHAP provide useful explainability capabilities, selecting an explainability mechanism requires balancing interpretability quality and explanation consistency. LIME generates local approximations that may vary depending on sampled observations, whereas SHAP provides theoretically consistent explanations derived from cooperative game theory principles. Consequently, SHAP was considered more suitable for FinSentinel because governance-aware environments require stable explanations capable of supporting both analyst trust and operational decision support (Ribeiro et al., 2016; Lundberg and Lee, 2017).

Figure 2.2: Explainable AI Workflow Using SHAP



*Figure 2.2 illustrates the process through which SHAP generates interpretable feature contributions from machine learning predictions.*

Table 2.2: Comparison of Explainability Methods

| Method | Advantages                                 | Limitations               | Suitability |
|--------|--------------------------------------------|---------------------------|-------------|
| LIME   | Local explanations, model-agnostic         | Explanation instability   | Moderate    |
| SHAP   | Local and global explanations, consistency | Higher computational cost | High        |

*Table 2.2 compares selected explainability approaches according to advantages, limitations, and suitability.*

## 2.4 Human–AI Collaboration in Fraud Detection

The increasing deployment of Artificial Intelligence within decision-making environments has created growing interest in Human–AI collaborative systems. Traditional AI implementations frequently aimed to automate operational processes entirely with minimal human intervention. Although fully autonomous systems can improve efficiency and reduce processing time, excessive dependence on automation may introduce risks associated with incorrect predictions, reduced accountability, and limited user trust. Within financial fraud detection environments, incorrect decisions may result in significant financial losses (Suresh and Guttag, 2021) or negative customer experiences, making complete automation potentially unsuitable (Amershi et al., 2019).

Human–AI collaboration attempts to combine the strengths of machine intelligence and human expertise within integrated operational workflows. Machine learning systems provide advantages including rapid data processing, pattern identification, and predictive capability, whereas human analysts contribute contextual understanding, domain knowledge, judgement, and ethical reasoning. Rather than replacing human decision-making entirely, collaborative systems utilise AI as a decision-support mechanism capable of assisting human users during operational processes.

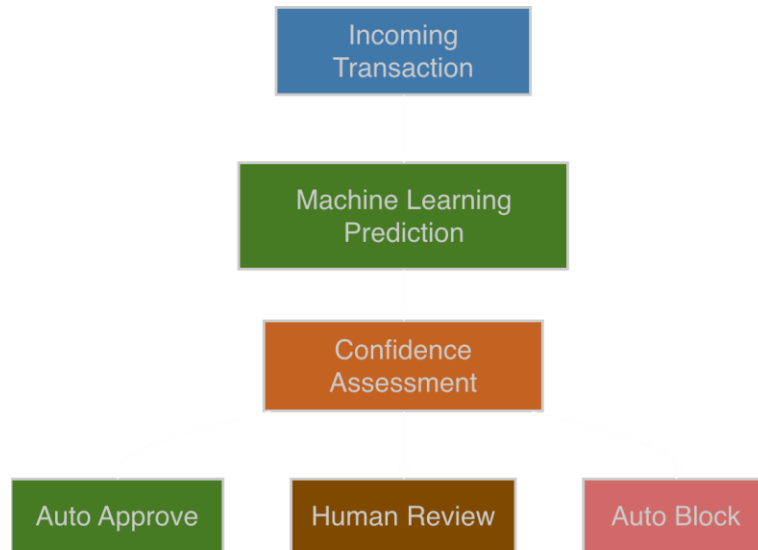
Several Human–AI approaches have been proposed within existing research. Human involvement may occur before model deployment through data preparation and training activities, during prediction generation through real-time intervention, or after predictions through verification and decision review processes. Human-in-the-Loop (HITL) systems represent one commonly adopted collaborative framework in which uncertain predictions are selectively escalated to human analysts for further investigation (Dellermann et al., 2019).

Within fraud detection environments, selective escalation mechanisms may provide important operational benefits. High-confidence predictions can be processed automatically, while transactions associated with greater uncertainty can be redirected for human review. Such approaches may reduce analyst workload while maintaining appropriate oversight of potentially risky transactions. Human intervention additionally improves transparency because analysts remain involved in critical decisions rather than relying entirely on automated systems.

The proposed FinSentinel framework adopts a Human–AI collaborative approach through governance-aware routing mechanisms that selectively assign transactions to automated approval, analyst review, or automated blocking categories. The framework therefore attempts to balance predictive efficiency with operational trust and accountability.

Consequently, results suggest that Human–AI collaboration may improve the reliability and practical applicability of fraud detection systems by combining computational intelligence with human judgement.

Figure 2.3: Human–AI Collaborative Fraud Detection Workflow



*Figure 2.3 illustrates a Human–AI collaborative workflow where prediction confidence influences routing decisions.*

## 2.5 Behavioural Drift Intelligence

Machine learning systems commonly assume that future observations follow patterns similar to historical training data. However, practical operational environments frequently experience changing behavioural characteristics due to evolving customer activities, emerging fraud strategies, and modifications in transaction patterns. When such changes occur, relationships previously learned by predictive models may gradually become less representative of current conditions. This phenomenon is generally referred to as **concept drift** or **behavioural drift** (Gama et al., 2014).

Concept drift occurs when the statistical properties of incoming data change over time, leading to a reduction in predictive reliability. Several forms of drift have been identified within existing literature. **Sudden drift** occurs when behavioural patterns change rapidly over a short period, while **gradual drift** develops through progressive modifications across longer periods.

**Incremental drift** reflects continuous slow changes, whereas **recurring drift** involves patterns that disappear and later re-emerge (Lu et al., 2018). Such changes can negatively affect predictive performance because machine learning models may continue operating based on outdated assumptions.

Traditional fraud detection systems frequently focus on predictive performance at the time of model development while providing limited support for continuous behavioural monitoring. Although models may initially achieve strong accuracy, performance degradation may occur if changing patterns remain undetected. Consequently, monitoring mechanisms have become increasingly important within deployed machine learning systems because they support the identification of emerging behavioural changes and improve long-term reliability.

Several approaches have been proposed for drift detection, including statistical monitoring techniques, window-based comparison methods, and sequential analysis approaches. One commonly used technique is the **Page–Hinkley algorithm**, which detects changes in data distributions by continuously monitoring cumulative deviations from expected values (Page, 1954). The method is particularly useful for identifying shifts in streaming environments because it can detect behavioural changes while reducing sensitivity to minor fluctuations.

Within the proposed FinSentinel framework, behavioural drift intelligence was incorporated as an additional monitoring mechanism designed to support governance recommendations and operational awareness. Drift information was not intended to replace predictive decision-making but rather to provide contextual intelligence capable of indicating whether behavioural changes may influence system reliability. This approach supports proactive monitoring and may assist analysts in identifying emerging risks within changing environments.

Consequently, behavioural drift monitoring represents an important capability for maintaining long-term reliability within machine learning systems because operational environments rarely remain static over extended periods.

## 2.6 AI Governance in Fraud Detection

The increasing adoption of Artificial Intelligence within operational environments has introduced growing concerns regarding accountability, transparency, fairness, and decision reliability. Although machine learning systems can provide substantial improvements in predictive performance, deploying such systems without appropriate control mechanisms may create operational and ethical risks. High-performing models may generate recommendations that are difficult to interpret, potentially affecting trust and limiting practical deployment within sensitive domains such as financial services (Jobin et al., 2019).

AI governance refers to the processes, principles, and mechanisms designed to ensure that Artificial Intelligence systems operate responsibly and consistently with organisational objectives. Governance mechanisms commonly focus on areas including transparency, explainability, risk management, human oversight, and continuous monitoring. Effective governance frameworks attempt to balance predictive capability with operational accountability by ensuring that AI systems remain understandable and controllable during practical deployment.

Several international initiatives have highlighted the importance of responsible AI implementation. Principles proposed by organisations including the OECD and emerging AI regulatory frameworks emphasise transparency, robustness, fairness, and human-centred decision-making (OECD, 2019). Within financial environments, governance becomes particularly important because automated decisions may directly influence customer experiences and organisational risk exposure.

Traditional fraud detection systems frequently concentrate on classification outcomes without providing mechanisms for operational control. Such systems may automatically approve or reject transactions solely based on prediction outputs, creating challenges when uncertain cases arise. Governance-aware systems attempt to address these limitations by introducing decision-routing mechanisms capable of incorporating confidence information, explainability outputs, and contextual intelligence into operational workflows.

Within the proposed FinSentinel framework, governance mechanisms were incorporated through threshold-based routing logic designed to classify transactions into automated approval, analyst review, or automated blocking categories. Governance threshold experimentation was additionally introduced to evaluate operational trade-offs between fraud capture effectiveness and analyst workload. This approach extends prediction-focused systems toward operationally controlled decision environments.

Consequently, AI governance represents an essential requirement within modern fraud detection systems because it supports transparency, improves accountability, and strengthens user trust.



## 2.7 Research Gap

Existing literature demonstrates significant progress in fraud detection through machine learning, explainability mechanisms, and Human–AI collaboration approaches. Numerous studies have focused on improving predictive performance using advanced machine learning algorithms, while others have investigated explainability methods and concept drift monitoring techniques. However, many existing studies examine these components independently rather than integrating them into a unified operational framework.

Although previous research has demonstrated the usefulness of explainable models and Human–AI collaboration, limited attention has been given to governance-aware environments where prediction outputs, human oversight, behavioural monitoring, and operational routing mechanisms function collectively. Furthermore, many studies primarily evaluate predictive performance metrics while providing limited analysis of operational effectiveness and decision-support processes.

The identified research gap therefore involves the absence of integrated fraud detection frameworks capable of combining predictive intelligence, explainability, governance mechanisms, Human–AI collaboration, behavioural drift monitoring, and operational effectiveness evaluation within a single environment.

The proposed FinSentinel framework attempts to address this limitation through the development of a governance-aware Human–AI collaborative system designed to improve transparency, trust, and long-term operational reliability.

Table 2.3: Literature Gap Analysis

| Study Area             | Existing Focus            | Limitation                     | Gap Addressed by FinSentinel |
|------------------------|---------------------------|--------------------------------|------------------------------|
| Fraud Detection        | Predictive accuracy       | Limited transparency           | Explainability integration   |
| Explainable AI         | Interpretation methods    | Limited operational usage      | Human decision support       |
| Human–AI Collaboration | Selective review          | Limited governance integration | Governance routing           |
| Drift Detection        | Behaviour monitoring      | Isolated implementation        | Integrated monitoring        |
| AI Governance          | Responsible AI principles | Limited fraud application      | Governance-aware framework   |

*Table 2.3 summarises research gaps identified within existing literature and corresponding contributions addressed by FinSentinel.*

# Methodology

## 3.1 Research Methodology

The project adopted an applied research methodology involving the design, implementation, and evaluation of a governance-aware Human–AI fraud detection framework.

A quantitative approach was employed because fraud detection performance can be assessed using measurable indicators including precision, recall, F1-score, AUC, and operational metrics.

A design-oriented methodology was additionally adopted because the framework required multiple integrated components including predictive intelligence, explainability, governance routing, calibration, behavioural monitoring, and dashboard interfaces.

## 3.2 Development Lifecycle

The development of FinSentinel followed an iterative Agile-inspired lifecycle to support incremental implementation and continuous refinement. An iterative approach was selected because the framework consisted of multiple interconnected modules where outputs from one component frequently influenced the design of subsequent components.

The process began with requirement analysis and literature investigation to identify limitations within existing fraud detection approaches. Following requirement identification, the transaction dataset underwent preprocessing procedures including cleaning, scaling, and preparation for machine learning implementation.

Machine learning models were subsequently developed and compared using performance measures suitable for imbalanced fraud detection problems. Following predictive model development, explainability functionality was integrated through SHAP analysis, while governance routing mechanisms were implemented to support selective decision escalation.

Behavioural drift monitoring, calibration analysis, and dashboard development were then progressively incorporated into the framework. Continuous testing and refinement activities ensured alignment between implementation decisions and project objectives.

Figure 3.1: FinSentinel Development Lifecycle



*Figure 3.1 illustrates the iterative development lifecycle adopted for FinSentinel, including requirement analysis, model development, integration, testing, and evaluation activities.*

### 3.3 Dataset Description

The proposed framework was developed using the publicly available Credit Card Fraud Detection dataset obtained from Kaggle (Kaggle, 2023). The dataset contains anonymised European card transaction records and is frequently used within fraud detection research because it represents realistic class imbalance conditions.

The dataset consists of **284,807 transaction records**, of which **284,315 represent legitimate transactions** and **492 represent fraudulent observations**. The dataset includes anonymised variables labelled V1–V28 generated using Principal Component Analysis (PCA), while the variables *Time* and *Amount* remain in their original form.

A major characteristic of the dataset involves severe class imbalance where fraudulent transactions represent approximately **0.173%** of all observations (He and Garcia, 2009). Such imbalance creates challenges because models may produce misleadingly high accuracy while failing to identify minority fraud instances effectively. Therefore, additional evaluation measures beyond overall accuracy were required.

Table 3.1: Dataset Characteristics

| Attribute               | Value            |
|-------------------------|------------------|
| Total Records           | 284,807          |
| Legitimate Transactions | 284,315          |
| Fraudulent Transactions | 492              |
| Features                | 30 + class label |
| Fraud Percentage        | 0.173%           |

*Table 3.1 summarises the key characteristics of the Credit Card Fraud Detection dataset used within the project.*

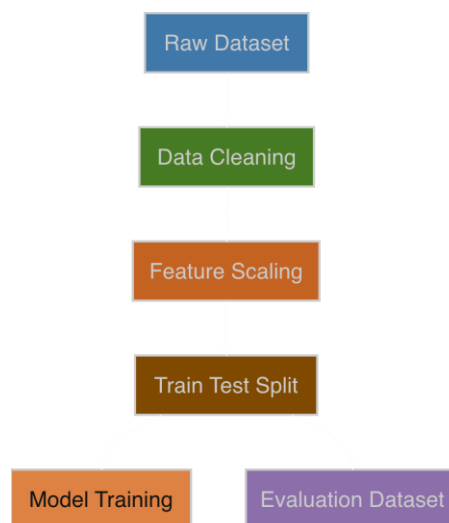
### 3.4 Data Preprocessing

Data preprocessing was performed to improve data quality and prepare observations for model development. Since machine learning performance may be negatively affected by inconsistent feature scales and imbalanced distributions, preprocessing represented an important stage within the implementation process.

Feature scaling was implemented using **StandardScaler**, which standardised numerical variables by transforming feature values toward zero mean and unit variance. Standardisation reduced the possibility of larger-value attributes disproportionately influencing model behaviour. The implementation code confirms the use of StandardScaler within the preprocessing pipeline.

The dataset additionally presented severe class imbalance because fraudulent transactions represented only a very small proportion of overall observations. Rather than using oversampling techniques, class imbalance handling was incorporated through balanced class weighting during model training. This approach reduced bias toward majority classes while avoiding potential overfitting risks associated with synthetic sampling methods. Preprocessed data was subsequently partitioned for model training and evaluation activities.

Figure 3.2: Data Preprocessing Pipeline



*Figure 3.2 illustrates the preprocessing workflow applied to transaction data before model training, including cleaning, feature scaling, and train-test partitioning.*

## 3.5 Model Selection Rationale

Multiple machine learning approaches were evaluated within the framework including Logistic Regression, Random Forest, XGBoost, LightGBM, CatBoost, and Multi-Layer Perceptron (MLP). Model comparison was performed because different algorithms exhibit varying strengths with respect to predictive capability, interpretability, and operational suitability.

Random Forest was selected as the principal model within FinSentinel because it demonstrated strong predictive capability while maintaining reasonable interpretability and robustness. Random Forest additionally provides advantages including reduced overfitting risk through ensemble learning and support for feature importance analysis.

Implementation settings further incorporated balanced class weighting to improve prediction behaviour within imbalanced conditions. The code implementation confirms the use of balanced class weighting during model construction.

The selected model therefore represented an appropriate balance between predictive performance and operational deployment requirements.

## 3.6 Governance Framework Design

The FinSentinel framework incorporated a governance layer designed to improve operational control and support Human–AI collaboration. Traditional fraud detection systems frequently rely solely on prediction outputs when generating decisions, which may introduce risks in situations involving prediction uncertainty. Therefore, an additional governance mechanism was introduced to improve decision reliability and operational trust.

The governance framework employed threshold-based routing logic to categorise transactions into operational decision groups. High-confidence legitimate transactions could be automatically approved, high-confidence fraud predictions could be automatically blocked, while uncertain predictions were escalated for human review. Rather than replacing predictive intelligence, governance mechanisms were designed to provide additional decision support.

To improve policy effectiveness, governance threshold experimentation was introduced to evaluate multiple threshold combinations and their influence on operational outcomes. Metrics including fraud capture rate, analyst workload, false positives, and automated decisions were monitored during experimentation. This enabled operational trade-offs to be evaluated and prevented decision-making from relying exclusively on predictive accuracy.

Governance experimentation was implemented as a methodology component, while resulting policy outcomes and threshold performance are presented and evaluated within Chapter 6.



### 3.7 Explainability Integration

Explainability functionality was integrated within the framework to improve transparency and support interpretation of prediction behaviour. Since machine learning models may operate as complex decision environments, prediction outcomes alone may not provide sufficient information for operational users.

SHAP (SHapley Additive Explanations) was selected because it provides mathematically grounded explanations capable of supporting both local and global interpretation. SHAP calculates feature contribution values according to cooperative game theory principles and determines the relative impact of individual variables on generated predictions (Lundberg and Lee, 2017).

Within FinSentinel, SHAP functionality was incorporated to support two forms of interpretation. Global interpretation enabled analysis of overall feature importance across the dataset, while local interpretation enabled investigation of individual transaction behaviour. This approach supported improved transparency and allowed users to understand factors influencing prediction outcomes.

### 3.8 Behavioural Drift Intelligence Integration

Behavioural drift monitoring functionality was integrated to improve long-term reliability and support operational awareness. Machine learning systems commonly assume stable data distributions; however, transaction characteristics may change over time because of evolving customer behaviour and emerging fraud strategies.

The proposed framework implemented behavioural monitoring using the **Page–Hinkley algorithm**, which continuously evaluates changes within incoming observations and identifies deviations from expected behaviour patterns. Prior to drift evaluation, transaction observations were ordered according to the Time attribute to preserve sequential behavioural characteristics. Temporal ordering was required because Page–Hinkley monitoring assumes progressive observation streams rather than randomly distributed samples. Temporal ordering was also preserved before data partitioning to better simulate operational deployment conditions and reduce the possibility of information leakage from future observations into historical training samples. This enabled behavioural variation to be evaluated in a manner more representative of practical transaction environments. Drift monitoring was incorporated as a supplementary intelligence mechanism rather than a replacement for predictive classification.

Drift outputs were transformed into contextual indicators including drift scores, severity classifications, and contributing drift factors. These outputs were intended to support operational users by providing awareness regarding possible behavioural changes that may influence prediction reliability.

## 3.9 Calibration Integration

Probability calibration was incorporated within FinSentinel to improve the reliability of prediction confidence values used by governance mechanisms. Raw prediction probabilities generated by machine learning models may not always accurately reflect observed outcomes. Consequently, high prediction confidence values do not necessarily indicate correspondingly high prediction reliability.

Calibration functionality was implemented using **CalibratedClassifierCV** with **sigmoid calibration (Platt Scaling)** (Platt, 1999) to improve probability estimation reliability (Niculescu-Mizil and Caruana, 2005). The code implementation confirms the use of calibrated classification methods during prediction generation.

Reliable confidence values were considered important because governance routing decisions depended partially on probability outputs. Improved calibration therefore strengthened the relationship between prediction confidence and operational decision-making processes.

## 3.10 Evaluation Metrics

Multiple evaluation measures were employed to assess predictive and operational performance within the proposed framework. Since fraud detection datasets frequently exhibit substantial class imbalance, overall accuracy alone was considered insufficient because high accuracy may be achieved despite poor fraud identification capability.

Predictive evaluation measures included **precision**, **recall**, **F1-score**, and **Area Under Curve (AUC)**. Precision measured the proportion of correctly identified fraudulent predictions among predicted fraud observations, while recall evaluated the ability to identify actual fraudulent transactions. F1-score provided a balanced measure combining precision and recall, while AUC evaluated overall discrimination capability.

Operational measures were additionally incorporated because framework performance extended beyond predictive accuracy. Metrics including fraud capture rate, analyst workload, human review frequency, automation rate, and trust indicators were evaluated to determine practical effectiveness.

Table 3.2: Evaluation Metrics

| Metric             | Purpose                              |
|--------------------|--------------------------------------|
| Precision          | Measure fraud prediction correctness |
| Recall             | Measure ability to detect fraud      |
| F1-score           | Balance precision and recall         |
| AUC                | Evaluate classification capability   |
| Fraud Capture Rate | Measure detected fraud proportion    |
| Human Review Rate  | Measure analyst workload             |
| Automation Rate    | Measure operational efficiency       |
| Brier Score        | Evaluate Probability Calibration     |

*Table 3.2 presents predictive and operational evaluation metrics used to assess framework performance.*

### 3.11 Tools and Technologies

Various tools and technologies were employed during implementation to support data processing, machine learning development, explainability integration, and dashboard construction. Implementation relied on established software libraries including Scikit-learn (Pedregosa et al., 2011), Pandas (McKinney, 2010), NumPy (Harris et al., 2020), Plotly (Plotly Technologies Inc., 2023), and Streamlit (Streamlit Inc., 2023).

Table 3.3: Tools and Technologies

| Tool         | Purpose                         |
|--------------|---------------------------------|
| Python       | Core development language       |
| Pandas       | Data manipulation               |
| NumPy        | Numerical operations            |
| Scikit-learn | Machine learning implementation |
| SHAP         | Explainability analysis         |
| Plotly       | Interactive visualisation       |
| Streamlit    | Dashboard development           |
| Page-Hinkley | Drift detection                 |
| Draw.io      | System diagrams                 |

*Table 3.3 summarises tools and technologies used during framework development and implementation.*

## 3.12 Verification and Validation

Verification and validation procedures were conducted throughout development to ensure that individual framework components operated correctly and aligned with project objectives. Verification focused on confirming that system components were implemented according to intended design specifications, while validation evaluated whether the developed framework successfully addressed the identified fraud detection problem.

Verification activities included preprocessing inspection, model configuration testing, governance threshold assessment, dashboard interaction checks, and consistency evaluation across interconnected modules. Validation activities focused on predictive and operational performance measures including ROC–AUC, PR–AUC, precision, recall, F1-score, fraud capture rate, and analyst workload indicators.

Explainability outputs were additionally inspected to ensure consistency between feature influence explanations and generated predictions, while behavioural monitoring functionality was evaluated using simulated behavioural variations to assess sensitivity toward changing operational conditions.

Table 3.4: Verification and Validation Activities Across FinSentinel Components

| Component          | Verification Method                  | Validation Method           |
|--------------------|--------------------------------------|-----------------------------|
| Data preprocessing | Scaling and missing value inspection | Dataset quality assessment  |
| Predictive model   | Hyperparameter testing               | Performance metrics         |
| Governance routing | Threshold inspection                 | Operational metrics         |
| Explainability     | SHAP consistency review              | Interpretation quality      |
| Drift monitoring   | Simulated drift testing              | Drift sensitivity           |
| Dashboard          | Interface functionality testing      | User interaction assessment |

*Table 3.4 summarises the verification and validation procedures conducted across major FinSentinel components to ensure implementation correctness and evaluate operational effectiveness.*

# System design and Architecture

## 4.1 Overall System Architecture

The FinSentinel framework was designed as a governance-aware Human–AI collaborative environment integrating predictive intelligence, explainability mechanisms, operational governance, behavioural monitoring, and interactive visualisation capabilities. Rather than functioning as an isolated prediction system, the framework was developed as an integrated decision-support environment capable of supporting both automated and human-assisted operational processes.

The architecture follows a modular design approach where each component performs specific responsibilities while remaining interconnected with other modules. Modular design principles were adopted because they improve maintainability, scalability, and flexibility by allowing components to be independently modified or enhanced without affecting overall system behaviour.

The architectural workflow begins with transaction input processing and data preparation activities before information is transferred toward predictive intelligence modules. Prediction outcomes subsequently flow through interconnected framework components that support operational decision-making and system monitoring.

Supporting system components were incorporated to enhance operational reliability and maintain system adaptability during deployment environments. Human interaction remains integrated throughout operational processes through governance-assisted escalation mechanisms.

Figure 4.1: FinSentinel Overall Architecture

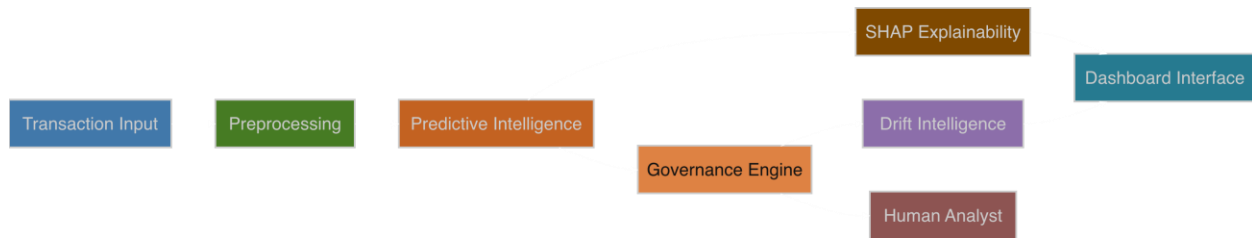


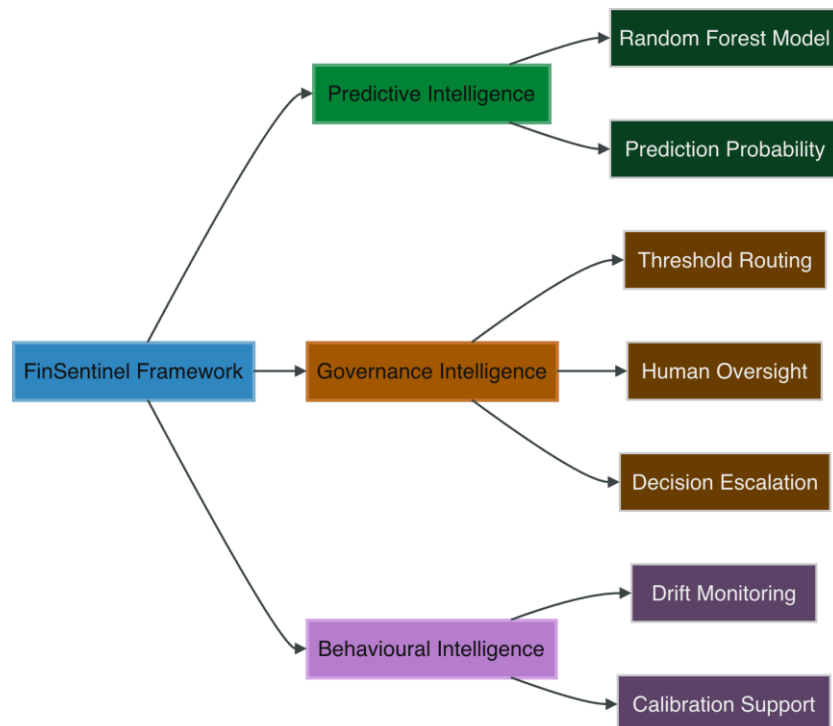
Figure 4.1 illustrates the high-level architecture of the proposed FinSentinel framework.

## 4.2 FinSentinel Framework Architecture

The framework architecture was designed around predictive intelligence and governance intelligence. Predictive intelligence identifies fraudulent behaviour using machine learning techniques, while governance intelligence supports routing decisions, confidence assessment, and behavioural monitoring.

Information generated across these components is integrated to improve transparency and operational decision support.

Figure 4.2: Dual Intelligence Architecture





*Figure 4.2 illustrates the relationship between predictive intelligence and governance intelligence within the FinSentinel framework.*

## 4.3 Human–AI Collaboration Pipeline

The proposed framework incorporates Human–AI collaboration through a selective escalation pipeline intended to balance automation efficiency with operational oversight. Rather than processing all transactions through fully autonomous mechanisms, decision pathways were designed according to prediction confidence and governance conditions.

High-confidence legitimate transactions are automatically approved, while high-confidence fraudulent transactions are automatically blocked. Predictions falling between governance threshold boundaries are redirected toward human analysts for further review. Additionally, behavioural drift indicators may influence routing behaviour by increasing human oversight during periods of detected instability.

This design approach attempts to reduce analyst workload while preserving human involvement for decisions associated with greater operational risk.

Human–AI collaboration within FinSentinel is achieved through governance routing mechanisms that selectively escalate uncertain transactions for analyst review while allowing confident decisions to be automatically processed.

Figure 4.3: Human–AI Collaboration Pipeline

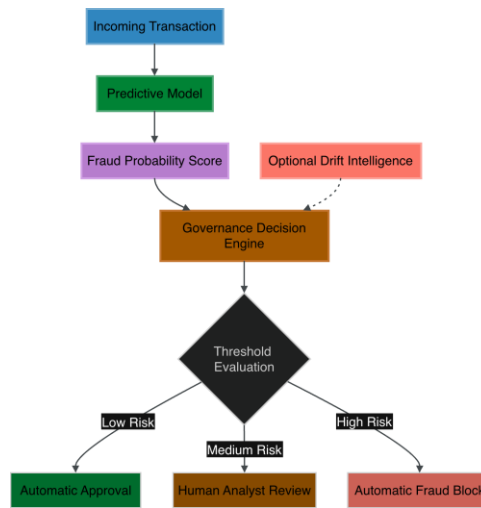


Figure 4.3 illustrates the selective escalation process where prediction confidence determines whether transactions are automatically processed or escalated for analyst review.

## 4.4 Governance Routing Engine

The governance routing engine was designed to provide an operational decision-support mechanism capable of translating prediction outputs into practical actions. Rather than relying solely on binary classification outcomes, the routing engine evaluates prediction confidence against configurable threshold boundaries to determine appropriate operational responses.

The routing process categorises predictions into multiple decision regions corresponding to automatic approval, human review, and automatic fraud blocking activities. Threshold boundaries can be modified according to different operational objectives including balanced governance, conservative governance, aggressive fraud containment, and operational efficiency strategies.

The governance mechanism therefore enables organisations to balance fraud capture effectiveness against operational workload while preserving transparency and human oversight within decision processes.

Figure 4.4: Governance Threshold Routing Logic

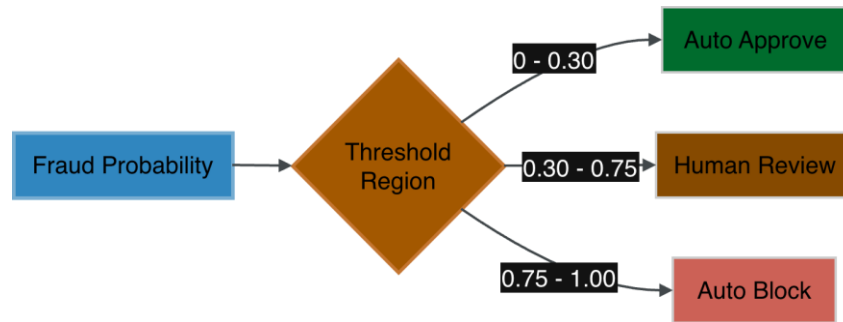


Figure 4.4 illustrates threshold-based governance routing used to convert prediction probabilities into operational decisions.

## 4.5 Dashboard Design Rationale

The FinSentinel dashboard was developed as a centralised decision-support interface designed to integrate predictive outputs, explainability information, governance indicators, behavioural monitoring results, calibration analysis, and operational performance metrics within a unified environment.

The interface structure was organised into modular components allowing users to navigate independently between predictive intelligence, governance threshold experimentation, explainability outputs, drift monitoring modules, and operational analytics functions. The modular architecture was selected to improve usability while reducing information fragmentation across system components.

Interactive visualisation mechanisms were additionally incorporated to improve interpretability and enable users to analyse relationships between prediction behaviour, governance outcomes, and operational performance indicators.

Figure 4.5: FinSentinel Dashboard Interface

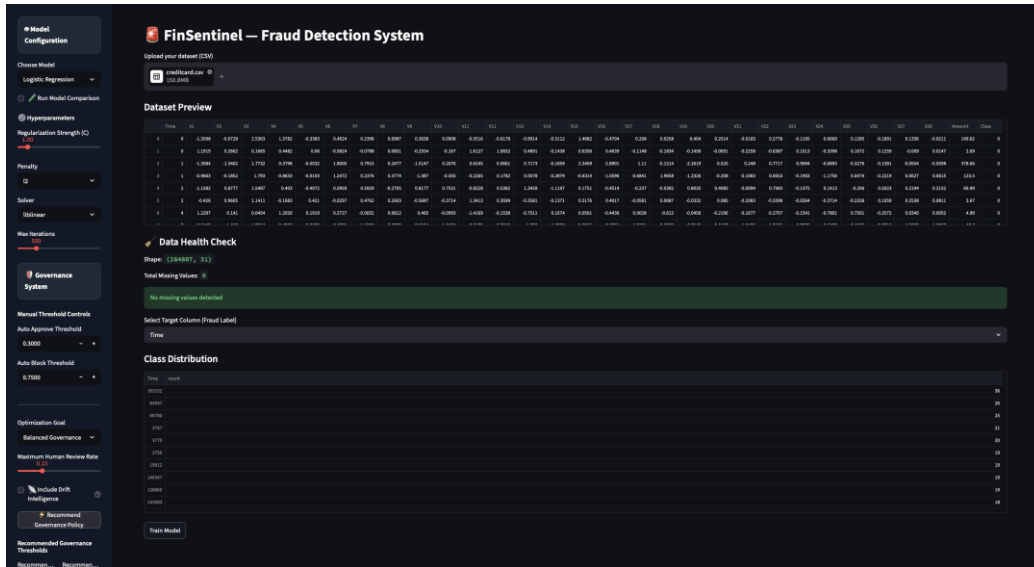


Figure 4.5: FinSentinel dashboard interface illustrating integrated predictive, governance, and behavioural monitoring modules

# Implementation

## 5.1 Technology Stack and Development Environment

FinSentinel was implemented using Python because of its extensive machine learning and data analysis support.

Streamlit was used for dashboard development, Scikit-learn for machine learning implementation, SHAP for explainability analysis, and Pandas, NumPy, Plotly, and Matplotlib for data processing and visualisation.

The modular implementation design enabled predictive, governance, and behavioural components to function independently while remaining integrated within the framework.

Figure 5.1: Technology Stack and Development Environment

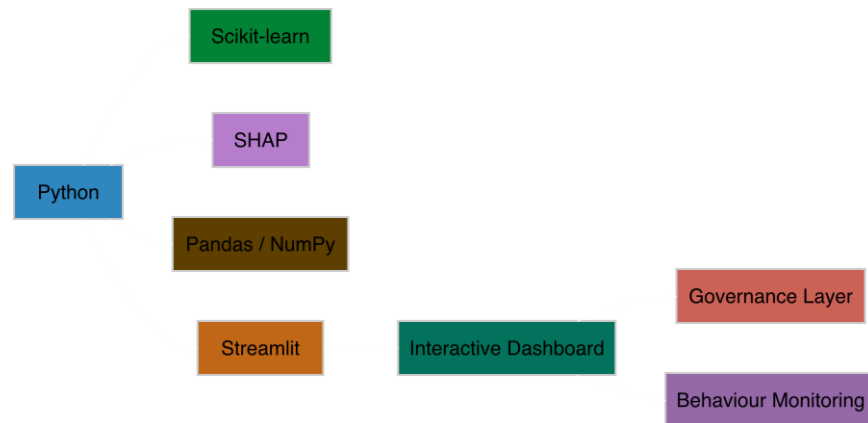


Figure 5.1 illustrates the technology stack and implementation environment adopted within FinSentinel.

## 5.2 Predictive Intelligence Implementation

The predictive intelligence module represented the primary analytical component within FinSentinel and was responsible for identifying potentially fraudulent transaction behaviour using supervised machine learning techniques. Multiple classification algorithms were implemented and comparatively evaluated to investigate differences in predictive capability and operational suitability.

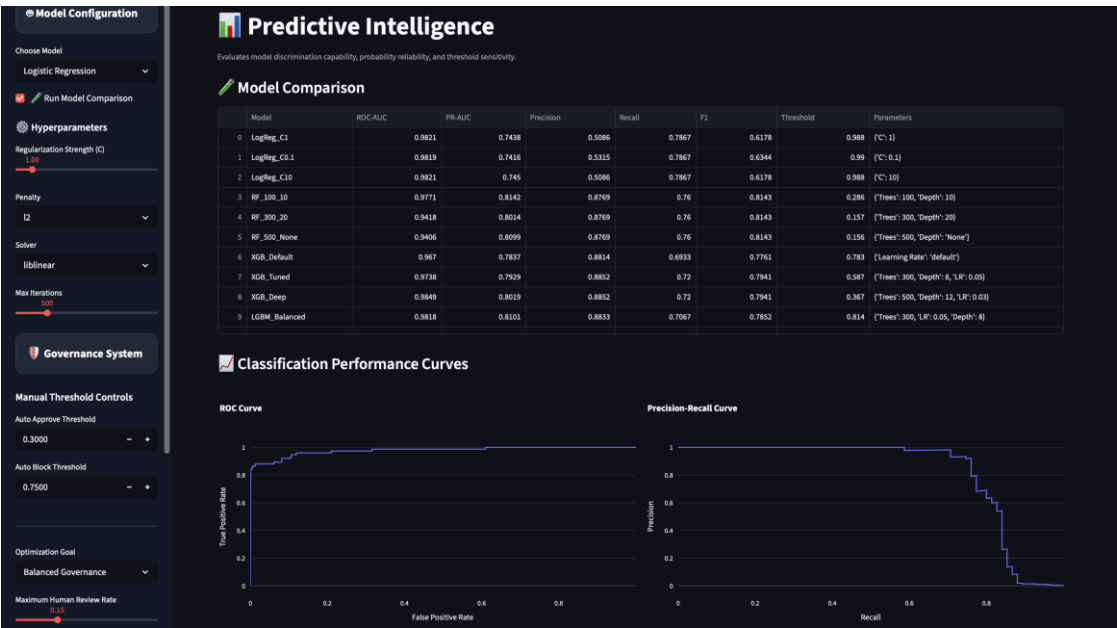
The implementation incorporated Logistic Regression, Random Forest, XGBoost, LightGBM, CatBoost, and Multi-Layer Perceptron models. Multiple experimental configurations involving different model parameters were evaluated to investigate variations in predictive and operational performance. Comparative experimentation enabled selection of suitable model configurations according to project requirements.

Comparative evaluation was performed using multiple performance indicators including ROC-AUC, PR-AUC, Precision, Recall, and F1-score measurements. Since fraud detection datasets exhibit significant class imbalance characteristics, evaluation extended beyond overall accuracy metrics to ensure reliable identification of minority fraud observations.

Experimental findings indicated that ensemble-based approaches demonstrated stronger predictive performance compared with simpler classification methods. Random Forest was subsequently selected as the primary operational model because it provided an effective balance between predictive capability, robustness, and interpretability.

Prediction outputs were integrated directly into downstream framework components including explainability mechanisms, governance routing functionality, calibration analysis, and behavioural monitoring systems. Prediction probabilities rather than binary outputs were utilised because governance mechanisms required confidence information for operational decision-making.

Figure 5.2: Comparative Performance Analysis of Machine Learning Models



|    | Model         | ROC-AUC | PR-AUC | Precision | Recall | F1     | Threshold | Parameters                                  |
|----|---------------|---------|--------|-----------|--------|--------|-----------|---------------------------------------------|
| 3  | RF_100_10     | 0.9771  | 0.8142 | 0.8769    | 0.76   | 0.8143 | 0.286     | ['Trees': 100, 'Depth': 10]                 |
| 5  | RF_500_None   | 0.9406  | 0.8099 | 0.8769    | 0.76   | 0.8143 | 0.156     | ['Trees': 500, 'Depth': 'None']             |
| 4  | RF_300_20     | 0.9418  | 0.8014 | 0.8769    | 0.76   | 0.8143 | 0.157     | ['Trees': 300, 'Depth': 20]                 |
| 8  | XGB_Deep      | 0.9849  | 0.8019 | 0.8852    | 0.72   | 0.7941 | 0.367     | ['Trees': 500, 'Depth': 12, 'LR': 0.03]     |
| 11 | LGBM_Deep     | 0.9775  | 0.7999 | 0.8852    | 0.72   | 0.7941 | 0.727     | ['Trees': 700, 'LR': 0.02, 'Depth': 15]     |
| 13 | CAT_Tuned     | 0.9638  | 0.7991 | 0.8852    | 0.72   | 0.7941 | 0.667     | ['Iterations': 500, 'Depth': 8]             |
| 7  | XGB_Tuned     | 0.9738  | 0.7929 | 0.8852    | 0.72   | 0.7941 | 0.587     | ['Trees': 300, 'Depth': 8, 'LR': 0.05]      |
| 14 | CAT_Deep      | 0.9786  | 0.809  | 0.8594    | 0.7333 | 0.7914 | 0.642     | ['Iterations': 700, 'Depth': 10]            |
| 9  | LGBM_Balanced | 0.9818  | 0.8101 | 0.8833    | 0.7067 | 0.7852 | 0.814     | ['Trees': 300, 'LR': 0.05, 'Depth': 8]      |
| 17 | MLP_Deep      | 0.9726  | 0.8188 | 0.98      | 0.6533 | 0.784  | 0.013     | ['Hidden': '(256,128,64)', 'Alpha': 0.0005] |
| 12 | CAT_Default   | 0.9486  | 0.7849 | 0.8235    | 0.7467 | 0.7832 | 0.491     | ['Mode': 'Default']                         |
| 10 | LGBM_Recall   | 0.9753  | 0.7993 | 0.8814    | 0.6933 | 0.7761 | 0.722     | ['Trees': 500, 'LR': 0.03, 'Depth': 12]     |
| 6  | XGB_Default   | 0.967   | 0.7837 | 0.8814    | 0.6933 | 0.7761 | 0.783     | ['Learning Rate': 'default']                |
| 15 | MLP_Default   | 0.9719  | 0.794  | 1         | 0.6267 | 0.7705 | 0.559     | ['Hidden': 100, 'Alpha': 0.0001]            |
| 16 | MLP_Tuned     | 0.9673  | 0.7821 | 0.9783    | 0.6    | 0.7438 | 0.014     | ['Hidden': '(128,64)', 'Alpha': 0.001]      |
| 1  | LogReg_C0.1   | 0.9819  | 0.7416 | 0.5315    | 0.7867 | 0.6344 | 0.99      | ['C': 0.1]                                  |
| 2  | LogReg_C10    | 0.9821  | 0.745  | 0.5086    | 0.7867 | 0.6178 | 0.988     | ['C': 10]                                   |
| 0  | LogReg_C1     | 0.9821  | 0.7438 | 0.5086    | 0.7867 | 0.6178 | 0.988     | ['C': 1]                                    |

Figure 5.2 illustrates comparative predictive performance across evaluated machine learning models.

Model ranking functionality was implemented to automatically compare experimental configurations according to multiple performance indicators and governance requirements. This mechanism enabled rapid identification of suitable predictive models prior to operational deployment.

Figure 5.3: Predictive Performance Visualisation Using ROC and Precision–Recall Curves

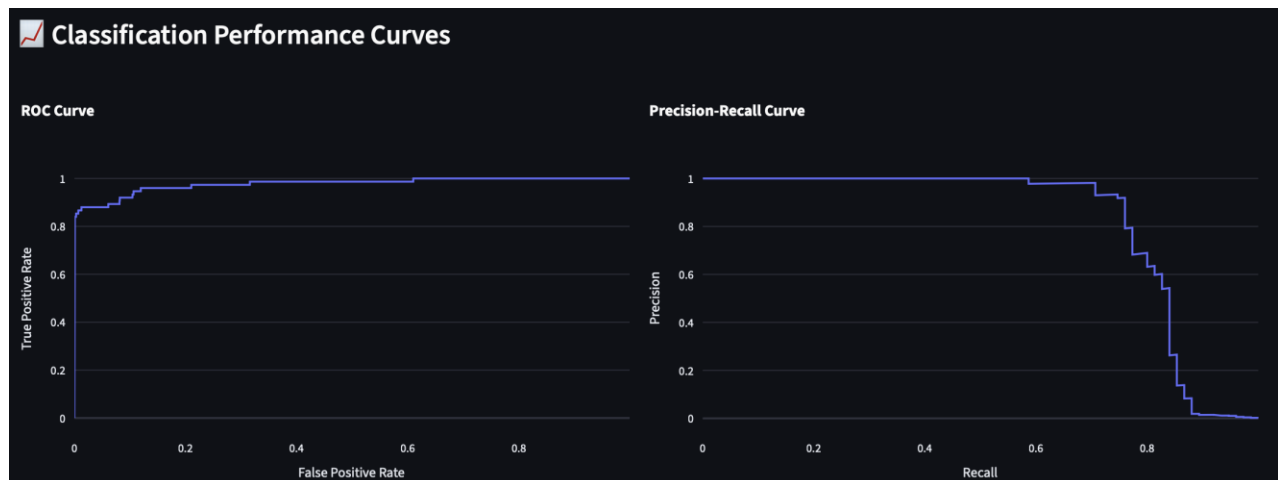


Figure 5.3 illustrates predictive performance through ROC and Precision–Recall analysis.

## 5.3 Explainability Intelligence Implementation

Explainability functionality was implemented within FinSentinel to improve transparency and enable users to understand factors influencing prediction behaviour. Since machine learning prediction outputs alone may not provide sufficient information regarding underlying decision processes, SHAP (SHapley Additive exPlanations) mechanisms were integrated into the dashboard environment.

The implementation supports both global and local interpretability mechanisms. Global explainability evaluates feature influence across overall model behaviour and provides an aggregated understanding of model decision patterns. Local explainability focuses on individual transaction predictions and enables investigation of factors responsible for specific prediction outcomes.

Global SHAP functionality was implemented to rank variables according to their contribution toward prediction generation. This allows users to identify dominant predictive characteristics and understand broader relationships learned by the model. Local SHAP analysis additionally enables transaction-level interpretation by identifying features that increase or decrease fraud probability for individual observations.

The explainability module was integrated directly into the dashboard environment to support interactive exploration of prediction reasoning. This functionality improves operational transparency and provides additional support for governance-assisted decision making.



Figure 5.4: Global SHAP Feature Importance Analysis

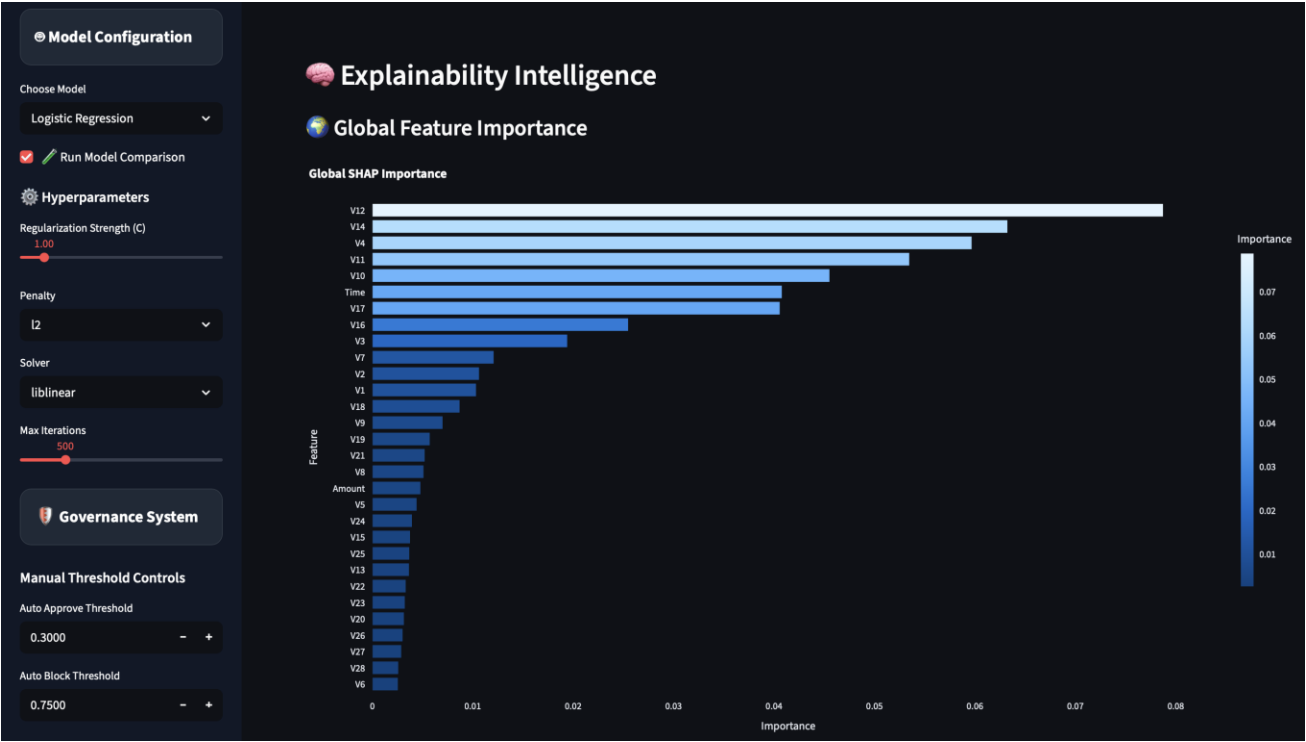


Figure 5.4 illustrates aggregated feature importance scores generated through SHAP analysis. Features exhibiting higher SHAP values contribute more significantly toward prediction behaviour across the overall dataset.

Figure 5.5: Local SHAP Transaction-Level Explanation.

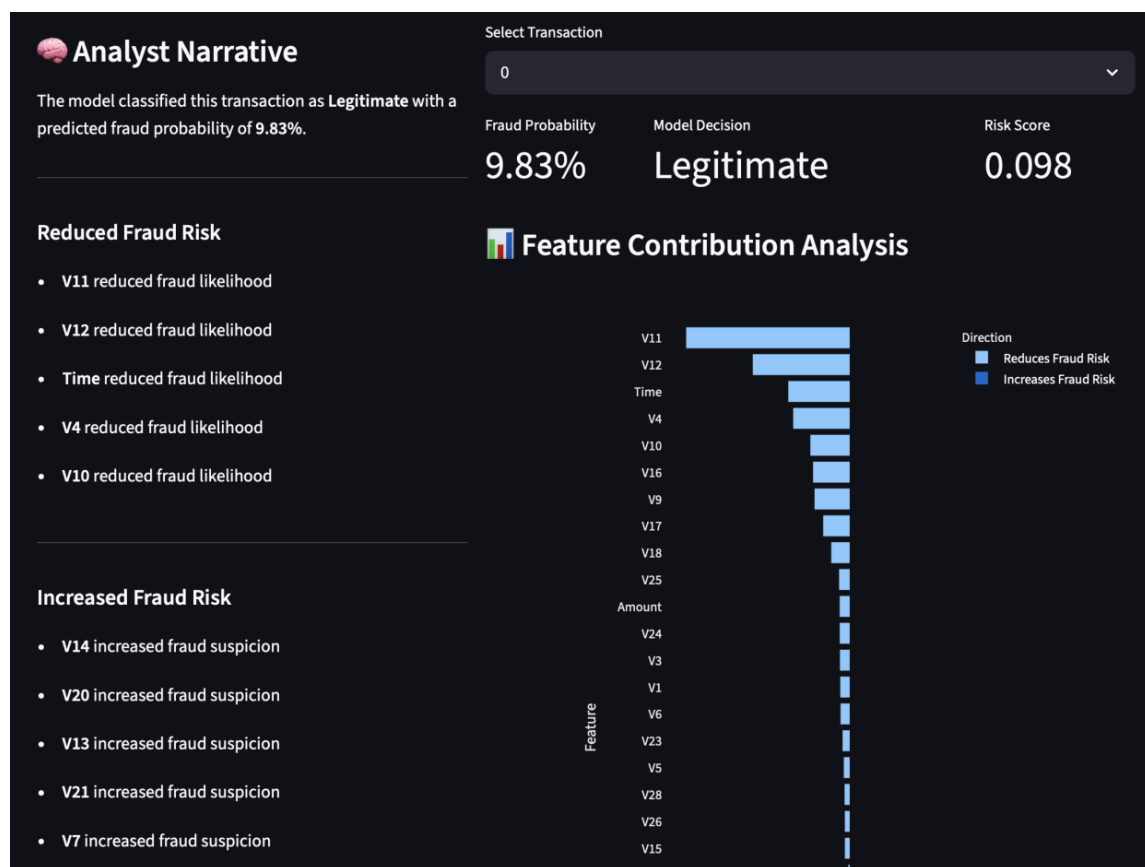


Figure 5.5 illustrates transaction-level explainability where feature contributions positively or negatively influence fraud prediction outcomes.

## 5.4 Governance System Implementation

The governance system was implemented as an operational routing layer designed to support Human–AI collaboration and improve decision reliability. Prediction probabilities generated by the Random Forest model were combined with governance thresholds to categorise transactions into automated approval, human review, and automated blocking groups.

Governance experimentation functionality was additionally implemented to evaluate the influence of different threshold combinations on operational behaviour. Users can modify approval and blocking thresholds and observe corresponding effects on fraud capture performance, automation rate, human review workload, and intervention outcomes. Multiple optimisation strategies including Balanced Governance, Aggressive Fraud Containment, Conservative Governance, and Operational Efficiency were implemented to simulate alternative operational policies and support intelligent recommendation of governance thresholds.

Operational metrics including fraud capture rate, human review rate, automation rate, and wrongful block rate were dynamically updated according to selected governance policies. This design enabled users to observe how policy modifications influenced operational outcomes.

Governance decisions generated by the routing engine were represented through operational performance indicators and routing visualisations. These outputs allow users to evaluate the practical implications of selected governance policies and understand the relationship between prediction behaviour and operational outcomes.

Figure 5.6: Operational Governance Performance Indicators

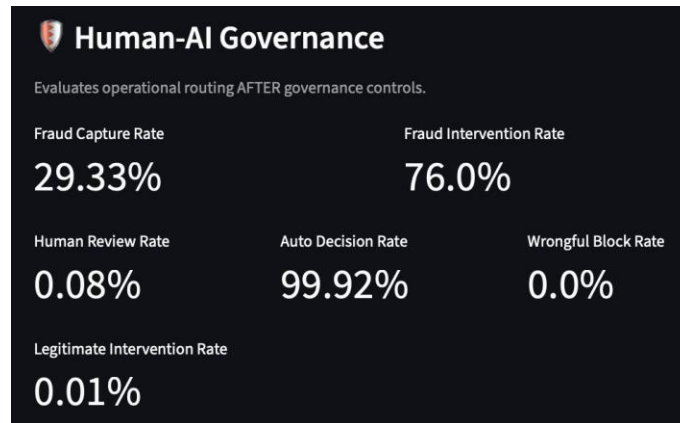
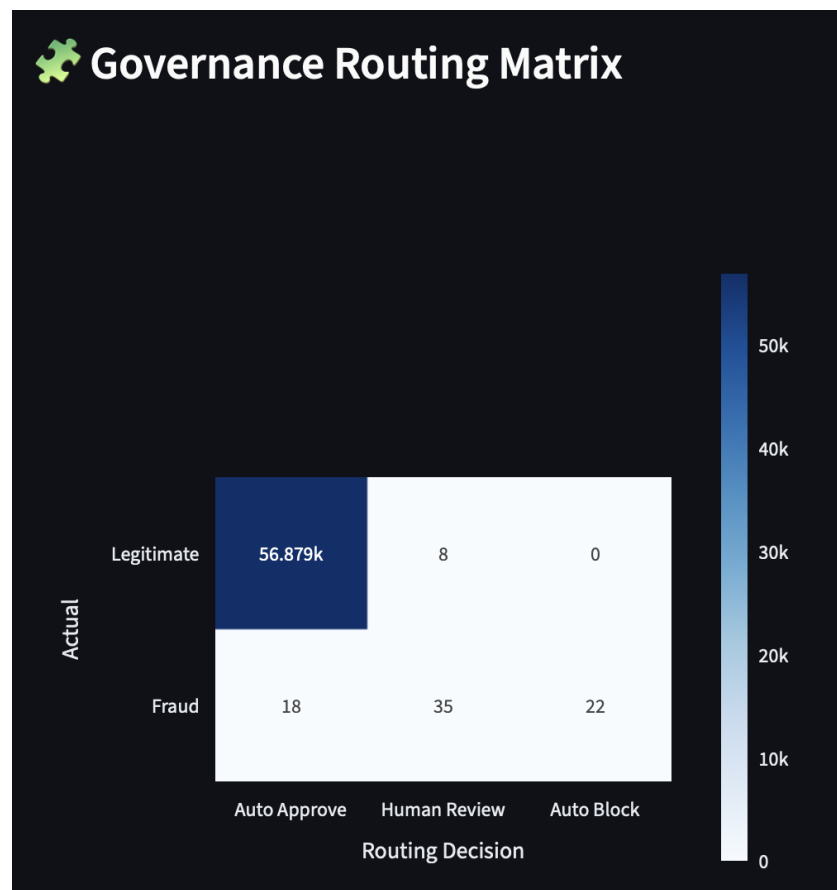


Figure 5.6 illustrates operational indicators generated following governance threshold application.

Figure 5.7: Transaction Routing Outcomes Across Governance Categories



*Figure 5.7 illustrates routing outcomes categorised into automatic approval, analyst review, and automatic fraud blocking.*

## 5.5 Governance Threshold Experimentation

Governance threshold experimentation functionality was implemented within FinSentinel to investigate the effect of operational decision boundaries on fraud detection behaviour. Rather than relying on static thresholds, the framework allows dynamic modification of approval and blocking boundaries to evaluate their influence on operational outcomes.

The experimentation environment supports multiple optimisation objectives including Balanced Governance, Aggressive Fraud Containment, Conservative Governance, and Operational Efficiency. These policies were designed to represent alternative operational priorities where users may favour stronger fraud interception, reduced analyst workload, or balanced decision performance.

Threshold recommendations were generated automatically using model performance measures and operational constraints. Users can additionally specify acceptable human review limits while optional behavioural drift information may be incorporated into threshold selection. This mechanism enables adaptive governance policies that respond to changing operational conditions.

Experimental outputs were dynamically updated to present metrics including fraud capture rate, automation rate, analyst review workload, and intervention outcomes. Such functionality supports comparative analysis of governance policies and assists identification of suitable operational configurations.

Figure 5.8: Threshold Sensitivity Analysis.

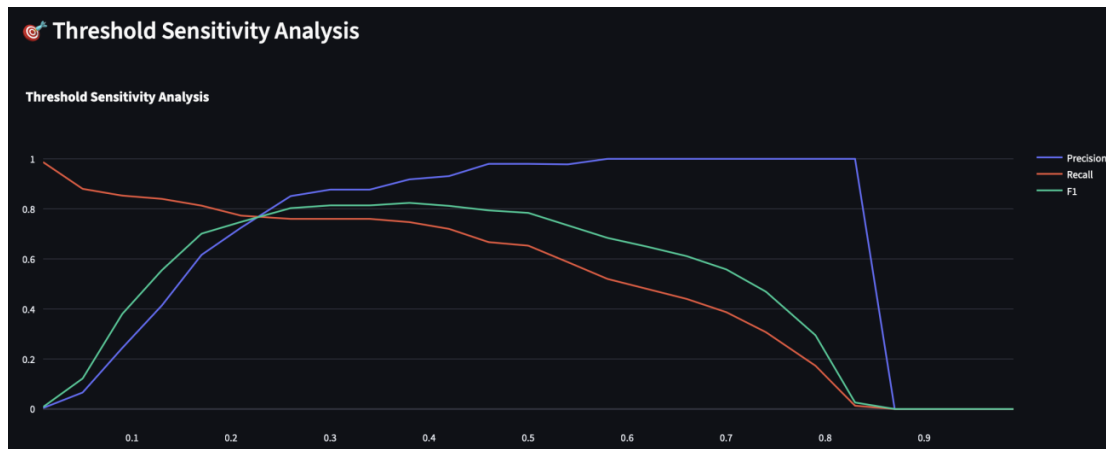


Figure 5.8 illustrates the relationship between precision, recall, and F1-score across varying threshold values.

Figure 5.9: Intelligent Governance Policy Recommendation Interface.

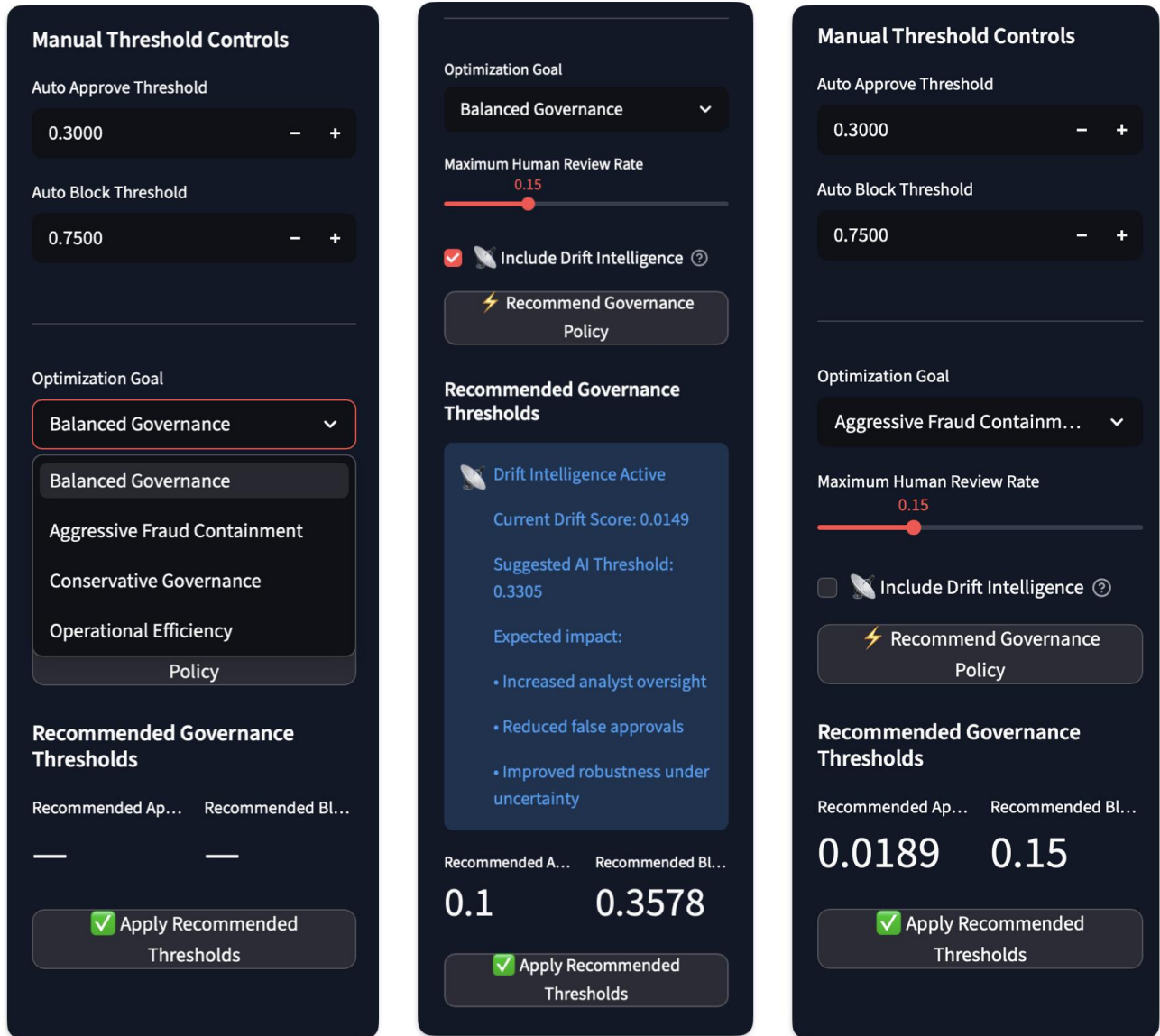


Figure 5.9 illustrates policy-based governance recommendations generated according to operational constraints and performance objectives.

Figure 5.10: Comparative Governance Threshold Experiment Results.

| ⚖️ Governance Threshold Experiment |         |       |               |                |             |              |                 |  |
|------------------------------------|---------|-------|---------------|----------------|-------------|--------------|-----------------|--|
|                                    | Approve | Block | Human Reviews | Auto Approvals | Auto Blocks | Capture Rate | False Positives |  |
| 0                                  | 0.05    | 0.5   | 991           | 55921          | 50          | 0.6533       | 1               |  |
| 1                                  | 0.05    | 0.55  | 998           | 55921          | 43          | 0.5733       | 0               |  |
| 2                                  | 0.05    | 0.6   | 1003          | 55921          | 38          | 0.5067       | 0               |  |
| 3                                  | 0.05    | 0.65  | 1008          | 55921          | 33          | 0.44         | 0               |  |
| 4                                  | 0.05    | 0.7   | 1012          | 55921          | 29          | 0.3867       | 0               |  |
| 5                                  | 0.05    | 0.75  | 1019          | 55921          | 22          | 0.2933       | 0               |  |
| 6                                  | 0.05    | 0.8   | 1034          | 55921          | 7           | 0.0933       | 0               |  |
| 7                                  | 0.05    | 0.85  | 1041          | 55921          | 0           | 0            | 0               |  |
| 8                                  | 0.05    | 0.9   | 1041          | 55921          | 0           | 0            | 0               |  |
| 9                                  | 0.1     | 0.5   | 186           | 56726          | 50          | 0.6533       | 1               |  |

Figure 5.10 illustrates operational outcomes across different governance threshold configurations.

Figure 5.11: Governance Policy Profile Analysis

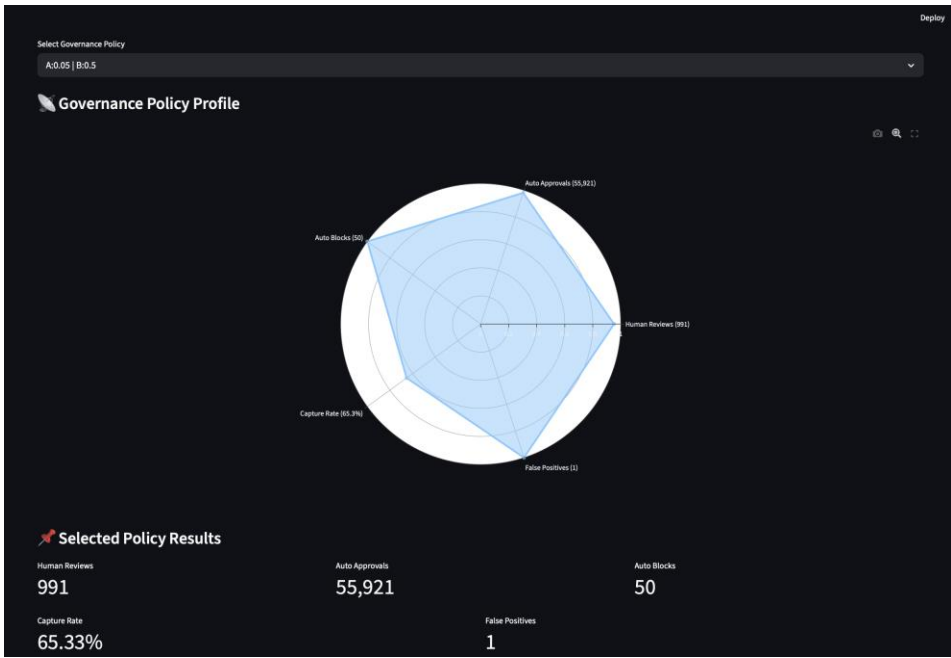


Figure 5.11 illustrates the selected governance policy profile and corresponding operational outcomes, including fraud capture rate, automated approvals, false positives, and routing behaviour metrics.



## 5.6 Behavioural Drift Intelligence Implementation

Behavioural drift intelligence was implemented within FinSentinel to improve awareness of evolving transaction characteristics and changing operational conditions. Since predictive models frequently assume stable underlying patterns, monitoring mechanisms were incorporated to identify changes capable of influencing model reliability and governance behaviour.

The implementation continuously evaluates transaction behaviour and transforms detected changes into interpretable operational indicators. Generated outputs include drift score measurements, drift severity classification, drift episode frequency, and average drift duration metrics. These indicators provide users with an overview of behavioural stability across operational environments.

Additional visualisation mechanisms were implemented to identify variables contributing most strongly toward detected behavioural changes. Drift driver analysis enables users to recognise recurring behavioural patterns and investigate features associated with operational variation.

Behavioural intelligence was further integrated with governance functionality to support adaptive policy selection. Drift information may optionally be incorporated into threshold recommendation processes, allowing operational decisions to consider changing environmental conditions.

Figure 5.12: Behavioural Drift Monitoring Indicators

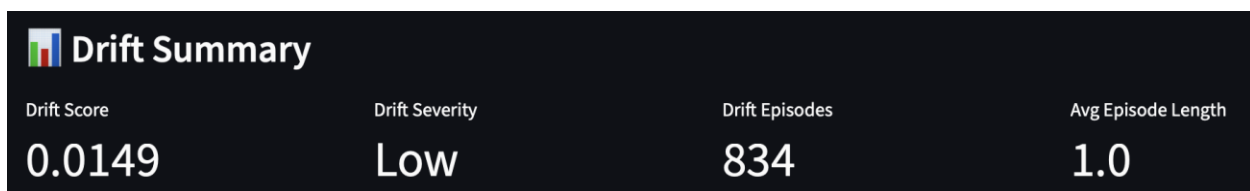


Figure 5.12 illustrates generated behavioural drift indicators including drift score, severity level, and drift statistics.

Figure 5.13: Feature-Level Behavioural Drift Contribution Analysis.

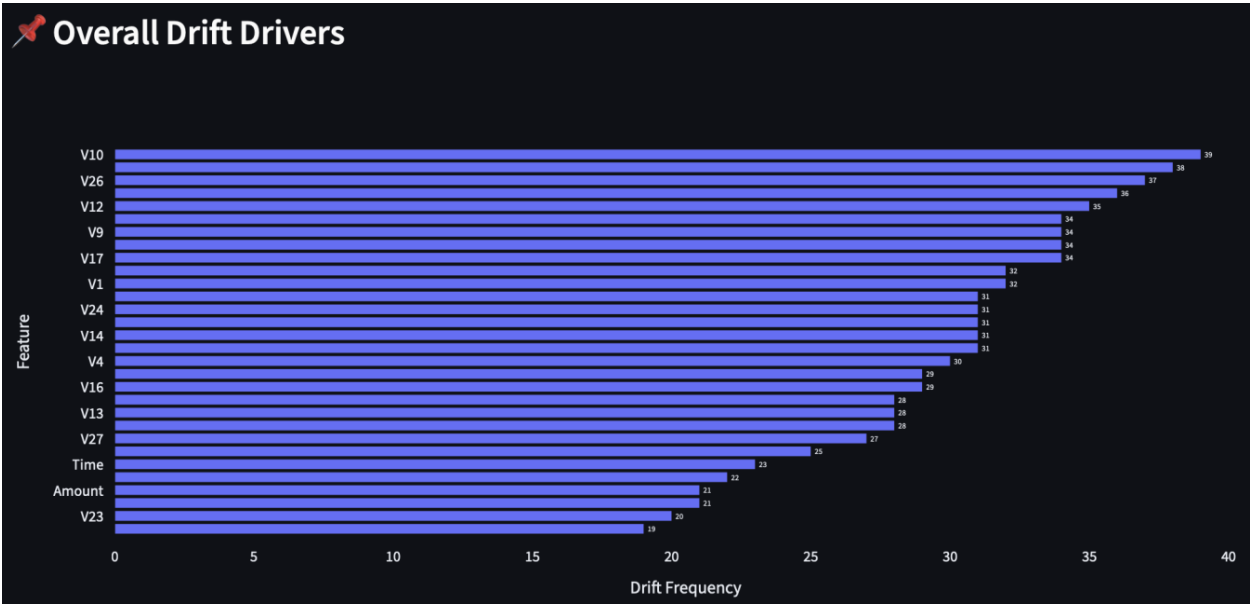


Figure 5.13 illustrates variables contributing most significantly toward detected behavioural drift.

Figure 5.14: Governance Behaviour During Drift Conditions.



Figure 5.14 illustrates the impact of behavioural drift on predictive and governance performance by comparing operational metrics before and after drift conditions.

## 5.7 Calibration Implementation

Calibration functionality was implemented within FinSentinel to improve the reliability of probability estimates generated by predictive models. Since governance routing decisions depend partially on confidence values, accurate probability estimation was considered important for operational decision-making.

Probability calibration was implemented using calibrated classification procedures to transform raw prediction outputs into probability estimates more closely aligned with observed outcomes. This process attempts to reduce inconsistencies between predicted confidence values and actual event frequencies.

Calibration functionality was incorporated primarily as an evaluation mechanism intended to assess the reliability of model confidence estimates. Although calibrated probabilities were available within the framework, the current implementation did not explicitly compare governance performance before and after calibration. Consequently, calibration results should be interpreted as supporting confidence stability rather than demonstrating direct operational improvement.

Figure 5.15: Probability Calibration Analysis



Figure 5.15 illustrates the relationship between predicted confidence values and observed outcomes.

## 5.8 Unified Transaction Intelligence Interface

The unified transaction intelligence interface was implemented to consolidate predictive outputs, governance decisions, and operational outcomes within a single analytical environment. The interface integrates actual transaction labels, model predictions, governance routing decisions, and fraud probability values to improve interpretability and support transaction-level investigation.

Filtering and search functionality were incorporated to enable exploration of transaction outcomes under different operational conditions. The integrated design reduces fragmentation between predictive and governance components and provides users with a complete view of decision behaviour.

Figure 5.16: Unified Transaction Intelligence Interface

The screenshot displays the 'Unified Transaction Intelligence Table' interface. At the top, there are four filter tabs: 'Actual Reality' (with 'Legitimate' and 'Fraud' options), 'AI Prediction' (with 'Legitimate' and 'Fraud' options), 'Governance Decision' (with 'Auto Approve', 'Auto Block', and 'Human Review' options), and 'Outcome' (with 'Correct Approval', 'Fraud Intercep...', and 'Missed Fraud' options). Below these filters is a 'Fraud Probability Range' slider set from 0.00 to 1.00. A 'Search Transaction ID' field is located above the table. The table itself has six columns: 'Transaction ID', 'Actual Reality', 'AI Prediction', 'Governance Decision', 'Fraud Probability', and 'Outcome'. It contains 19 rows of data, all showing 'Legitimate' for both Actual Reality and AI Prediction, 'Auto Approve' for Governance Decision, and 'Correct Approval' for Outcome. The Fraud Probability values range from 0.0001 to 0.0045.

| Transaction ID | Actual Reality | AI Prediction | Governance Decision | Fraud Probability | Outcome          |
|----------------|----------------|---------------|---------------------|-------------------|------------------|
| 0              | Legitimate     | Legitimate    | Auto Approve        | 0.0001            | Correct Approval |
| 1              | Legitimate     | Legitimate    | Auto Approve        | 0.0074            | Correct Approval |
| 2              | Legitimate     | Legitimate    | Auto Approve        | 0.0038            | Correct Approval |
| 3              | Legitimate     | Legitimate    | Auto Approve        | 0.0041            | Correct Approval |
| 4              | Legitimate     | Legitimate    | Auto Approve        | 0.072             | Correct Approval |
| 5              | Legitimate     | Legitimate    | Auto Approve        | 0.017             | Correct Approval |
| 6              | Legitimate     | Legitimate    | Auto Approve        | 0.0048            | Correct Approval |
| 7              | Legitimate     | Legitimate    | Auto Approve        | 0.0119            | Correct Approval |
| 8              | Legitimate     | Legitimate    | Auto Approve        | 0.0026            | Correct Approval |
| 9              | Legitimate     | Legitimate    | Auto Approve        | 0.0001            | Correct Approval |
| 10             | Legitimate     | Legitimate    | Auto Approve        | 0.0004            | Correct Approval |
| 11             | Legitimate     | Legitimate    | Auto Approve        | 0.0043            | Correct Approval |
| 12             | Legitimate     | Legitimate    | Auto Approve        | 0.0049            | Correct Approval |
| 13             | Legitimate     | Legitimate    | Auto Approve        | 0.0068            | Correct Approval |
| 14             | Legitimate     | Legitimate    | Auto Approve        | 0.0073            | Correct Approval |
| 15             | Legitimate     | Legitimate    | Auto Approve        | 0.0225            | Correct Approval |
| 16             | Legitimate     | Legitimate    | Auto Approve        | 0.0018            | Correct Approval |
| 17             | Legitimate     | Legitimate    | Auto Approve        | 0.0045            | Correct Approval |
| 18             | Legitimate     | Legitimate    | Auto Approve        | 0.0027            | Correct Approval |

Figure 5.16 illustrates the unified transaction intelligence environment integrating predictions, governance decisions, and transaction analysis.

# Results and Evaluation

## 6.1 Predictive Model Performance Evaluation

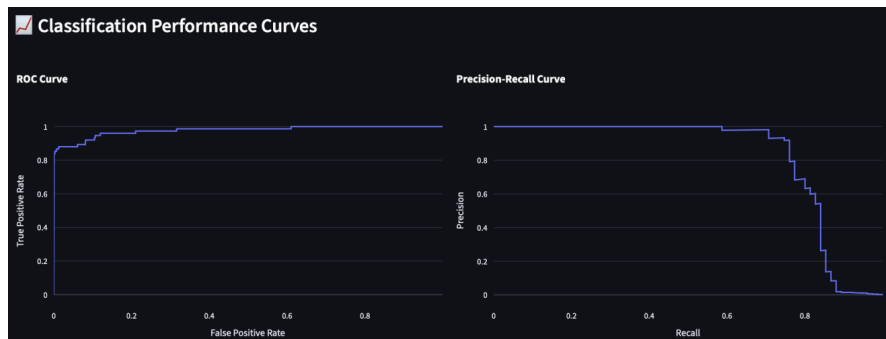
The predictive performance of FinSentinel was evaluated using multiple classification measures suitable for imbalanced fraud detection environments. Since fraudulent transactions represented only a small proportion of total observations, evaluation relied on precision, recall, F1-score, ROC–AUC, and Precision–Recall AUC rather than accuracy alone.

The achieved ROC–AUC value of 0.977 demonstrates strong discrimination capability between legitimate and fraudulent transactions across varying thresholds. However, ROC-based measures may occasionally provide optimistic interpretations in highly imbalanced datasets because true negatives dominate overall observations. Consequently, PR–AUC becomes particularly important because it focuses specifically on minority-class detection performance. The achieved PR–AUC value of 0.814 therefore provides stronger evidence regarding the capability of the model to identify rare fraudulent transactions reliably (Dal Pozzolo et al., 2015).

The Precision–Recall curve provides additional insight for imbalanced classification tasks because it focuses specifically on minority-class detection performance. The observed results indicate that the model maintained a balanced relationship between precision and recall, suggesting that fraudulent transactions were detected without excessive false positive generation.

Although predictive performance was strong, some degree of classification uncertainty remained because fraudulent behavioural patterns continuously evolve over time. Consequently, predictive outputs alone were considered insufficient for reliable operational deployment and additional explainability and governance mechanisms were integrated within the framework.

Figure 6.1: Predictive Performance Evaluation Using ROC and Precision–Recall Analysis



*Figure 6.1 illustrates predictive performance of the selected model using ROC and Precision–Recall analysis.*

## 6.2 Classification Error Analysis

Classification error analysis was performed to examine the distribution of prediction outcomes and identify potential sources of model error. Confusion matrix evaluation provides additional insight regarding false positives and false negatives that may influence operational decision-making.

Results indicate that most legitimate and fraudulent transactions were correctly classified; however, a limited number of misclassifications remained. False positives may increase analyst workload through unnecessary transaction investigations, whereas false negatives represent greater operational risk because fraudulent activity may remain undetected.

Consequently, subsequent governance mechanisms and Human–AI review processes were incorporated to reduce the impact of prediction uncertainty.

Figure 6.2: Confusion matrix for transaction classification performance.

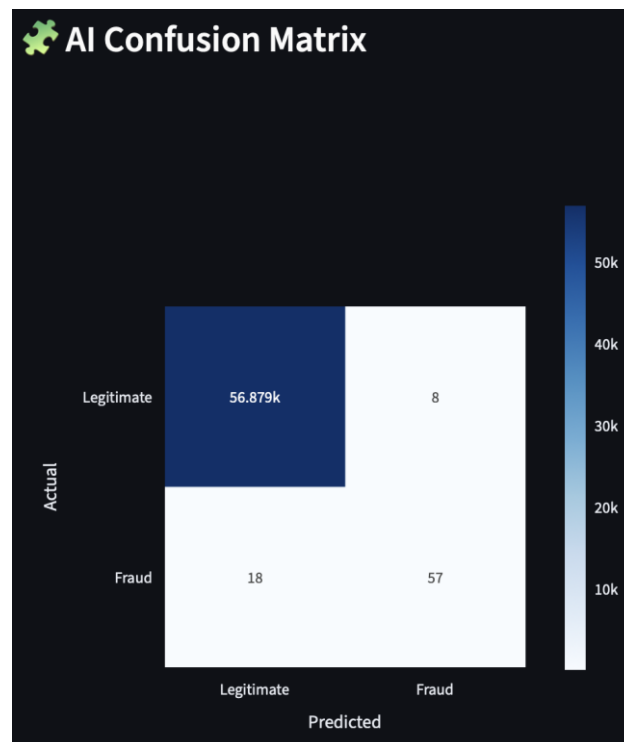


Figure 6.2 illustrates classification outcomes including true positives, true negatives, false positives, and false negatives.

## 6.3 Explainability Evaluation

Explainability analysis was performed to investigate variables contributing significantly toward prediction behaviour and evaluate the transparency of model decisions. Since predictive outputs alone provide limited insight regarding model reasoning, SHAP functionality was assessed to determine whether influential variables and transaction-level explanations could be effectively interpreted.

Global interpretation results indicated that features including V12, V14, V4, V11, and V10 demonstrated substantial influence on prediction outcomes. These findings suggest that fraud classification decisions were influenced by combinations of transformed transactional characteristics rather than isolated variables.

Local transaction-level interpretation additionally provided contextual explanations for individual prediction outcomes. The analyst narrative interface identified variables contributing positively and negatively toward fraud probability and translated model behaviour into human-readable explanations.

The results demonstrate that explainability functionality improved transparency while preserving predictive capability and strengthened Human–AI interaction by enabling users to understand both model-level and transaction-level reasoning.



Figure 6.3: Global feature influence identified using SHAP analysis.

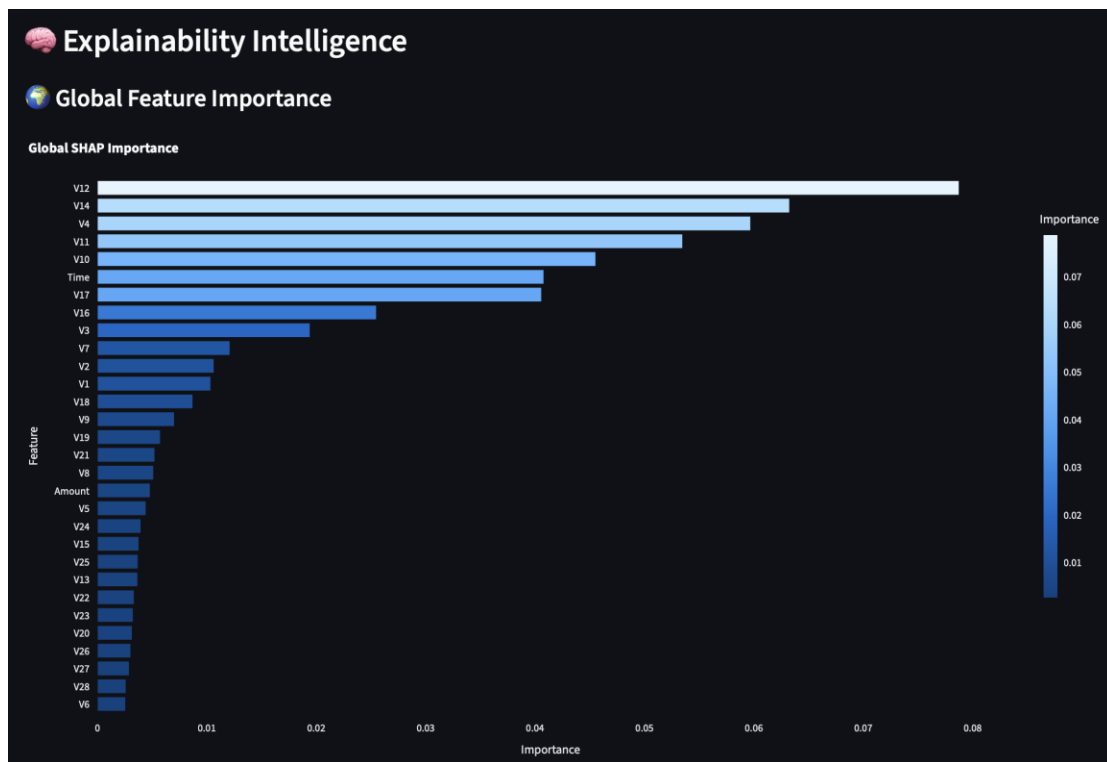


Figure 6.3 illustrates variables demonstrating the greatest influence on prediction behaviour.

Figure 6.4: Transaction-level explanation and analyst narrative output.

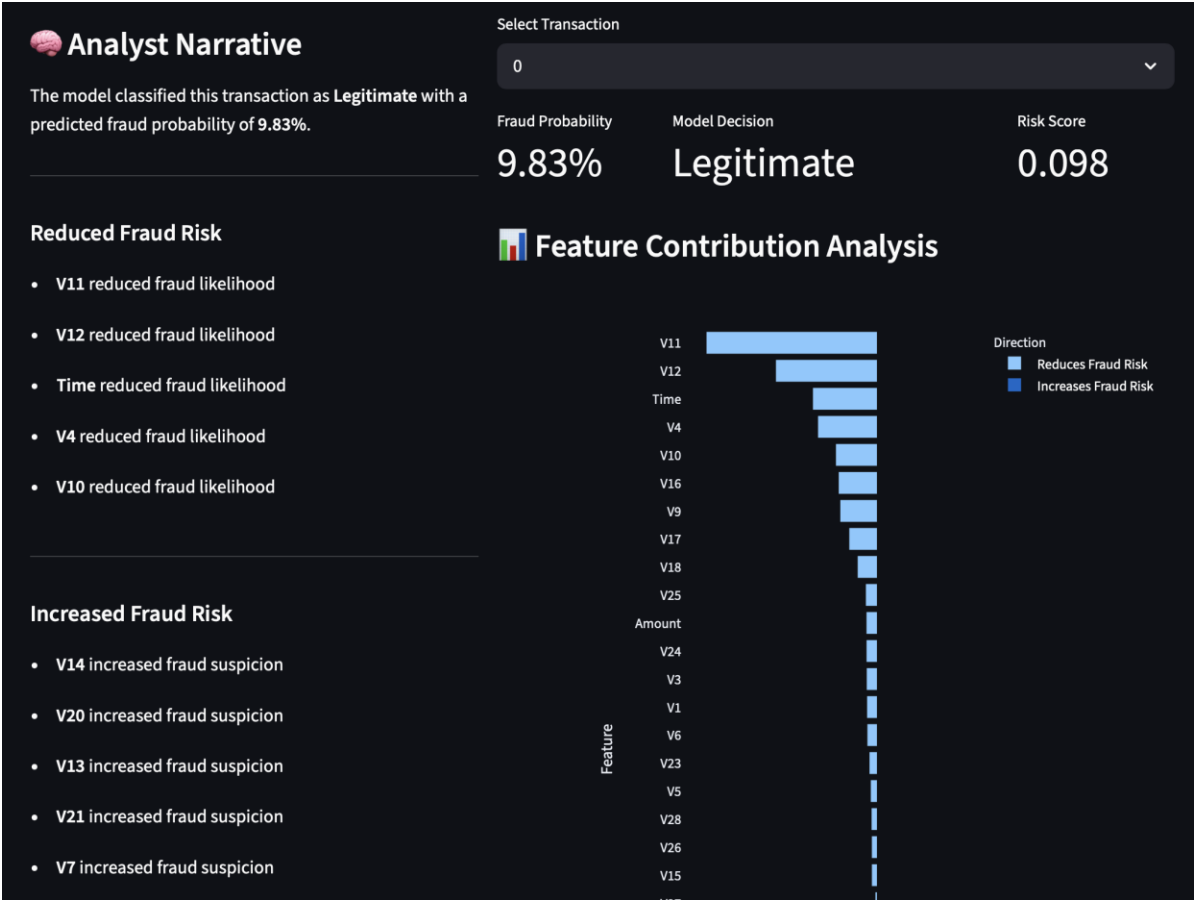


Figure 6.4 illustrates transaction-level prediction explanations and associated analyst interpretation outputs.

## 6.4 Governance Evaluation

Governance evaluation was performed to investigate operational trade-offs between fraud detection effectiveness and human intervention requirements. Multiple governance configurations including Balanced Governance, Aggressive Fraud Containment, Conservative Governance, and Operational Efficiency strategies were evaluated.

Results demonstrated substantial variation across governance policies. Aggressive Fraud Containment achieved the highest fraud capture rate (84.0%) but required significantly increased human review activity and generated a higher wrongful intervention rate. In contrast, Operational Efficiency maximised automation performance through a 99.86% automatic decision rate while reducing analyst involvement, although this occurred with lower fraud detection capability.

Balanced Governance demonstrated a compromise between operational efficiency and fraud detection performance by maintaining moderate intervention rates while preserving fraud capture capability. The results indicate that governance performance should not be evaluated solely through predictive accuracy because operational outcomes vary substantially depending on policy selection.

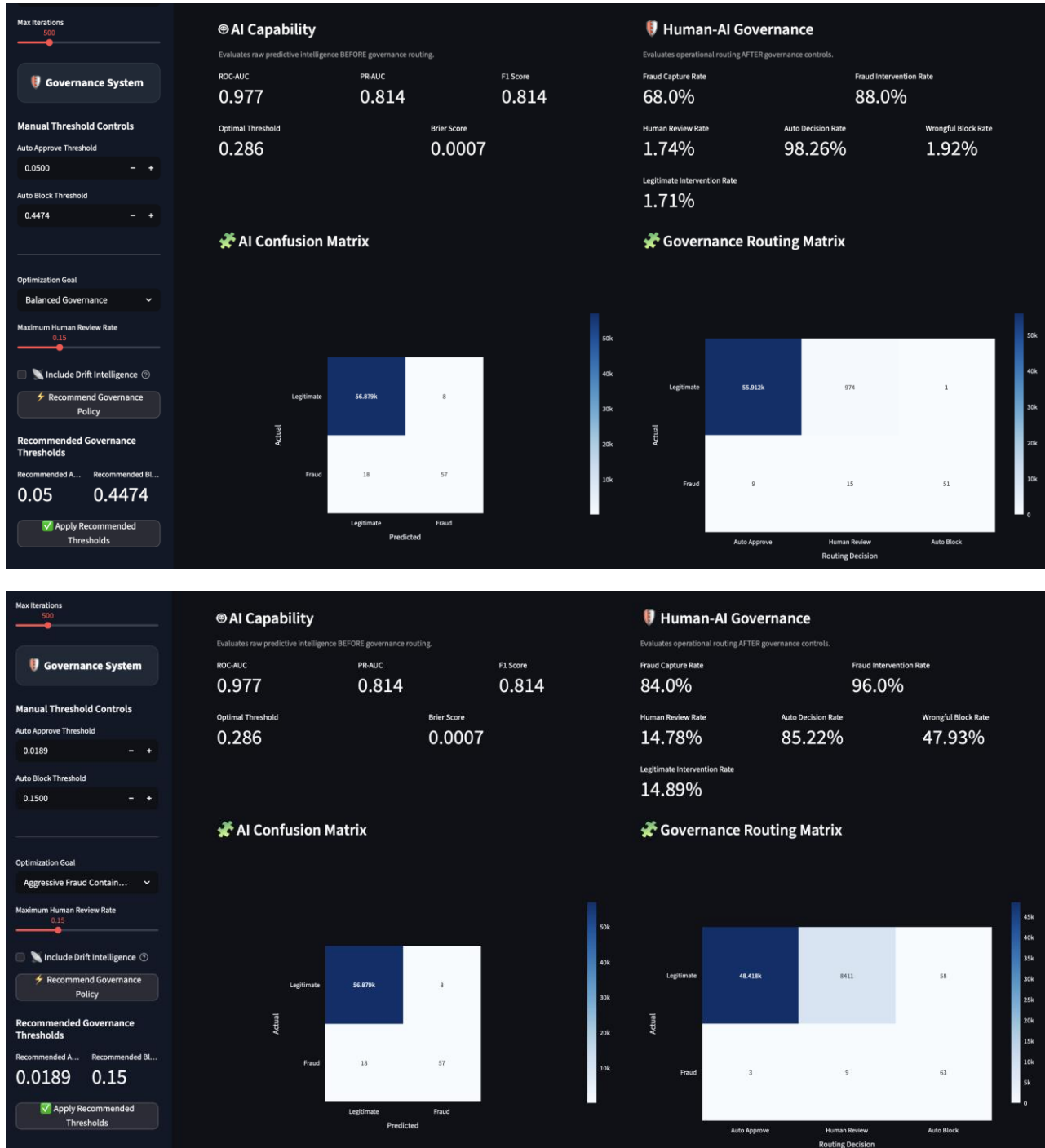
These findings suggest that governance mechanisms provide practical decision-support functionality by enabling organisations to select routing strategies according to operational priorities and risk tolerance. Although Aggressive Fraud Containment achieved the highest fraud capture capability, the associated increase in analyst workload and wrongful intervention rates suggests that maximising predictive performance alone may not necessarily represent an optimal operational strategy. This observation demonstrates that improving technical performance may introduce practical trade-offs involving operational cost, user experience, and human intervention requirements (Amershi et al., 2019).

Table 6.1: Governance Policy Comparison

| Policy                       | Fraud Capture | Human Review | Auto Decision | Wrongful Block |
|------------------------------|---------------|--------------|---------------|----------------|
| Aggressive Fraud Containment | 84.0%         | 14.78%       | 85.22%        | 47.93%         |
| Conservative Governance      | 56.0%         | 14.92%       | 85.08%        | 0.0%           |
| Operational Efficiency       | 56.0%         | 0.14%        | 99.86%        | 0.0%           |
| Balanced Governance          | 68.0%         | 1.74%        | 98.26%        | 1.92%          |

*Table 6.1 presents comparative operational outcomes across different governance policies.*

Figure 6.5: Governance Policy Comparison Results



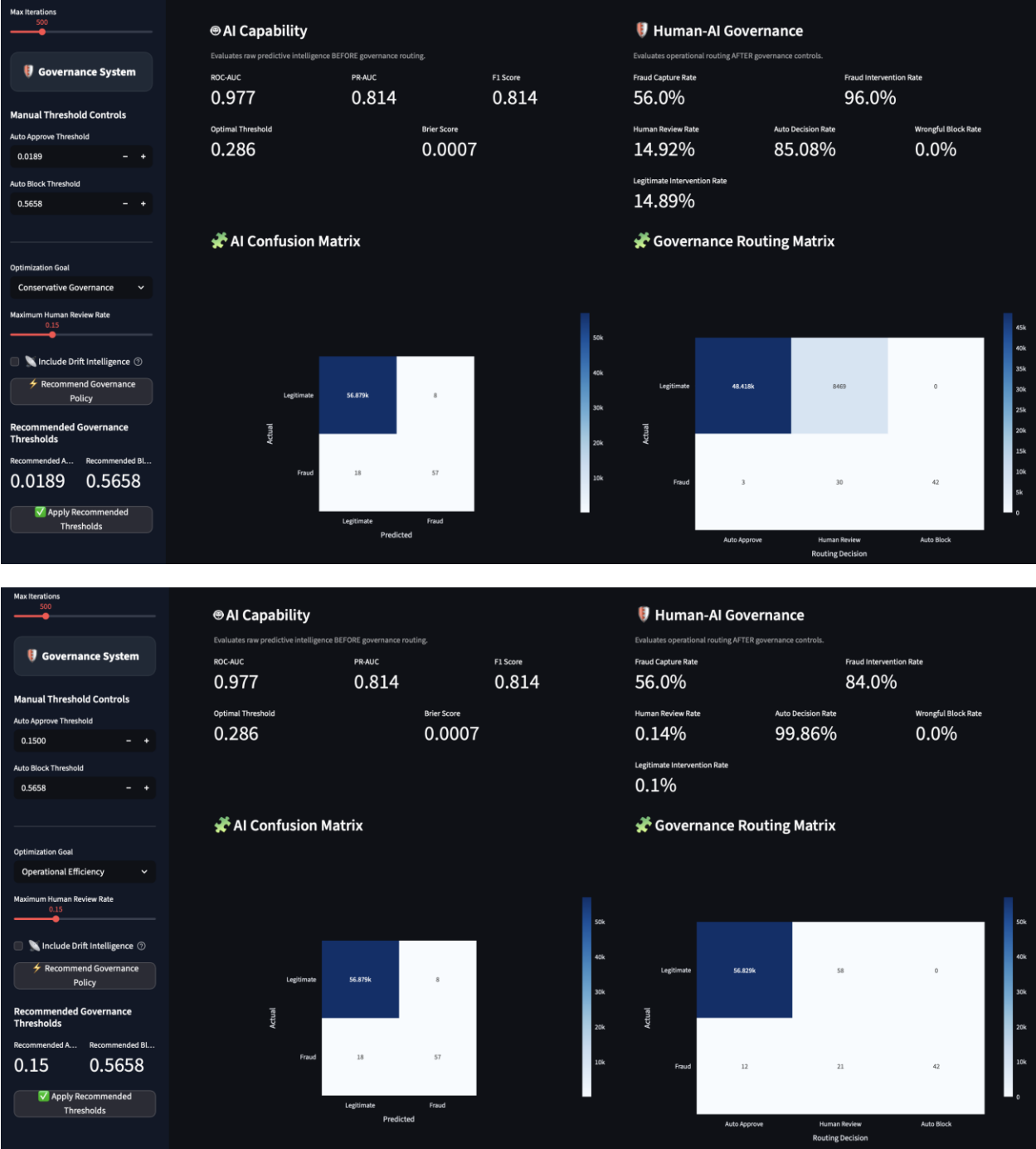


Figure 6.5 illustrates comparative operational outcomes across different governance strategies.

## 6.5 Threshold Sensitivity Evaluation

Threshold sensitivity evaluation was performed to investigate the relationship between classification thresholds and predictive performance measures. Since governance routing decisions depend directly on prediction confidence values, understanding threshold behaviour was considered important for operational decision-making.

Results demonstrated varying relationships among precision, recall, and F1-score as threshold values changed. Increasing threshold values generally improved precision while simultaneously reducing recall, indicating stricter prediction behaviour. Lower thresholds increased fraud detection capability but also increased the likelihood of false positive generation.

The findings highlight the importance of threshold selection because operational performance cannot be optimised through predictive accuracy alone. Appropriate threshold selection therefore requires balancing fraud detection effectiveness against analyst workload and intervention requirements.

Figure 6.6: Threshold sensitivity analysis showing precision–recall trade-offs.

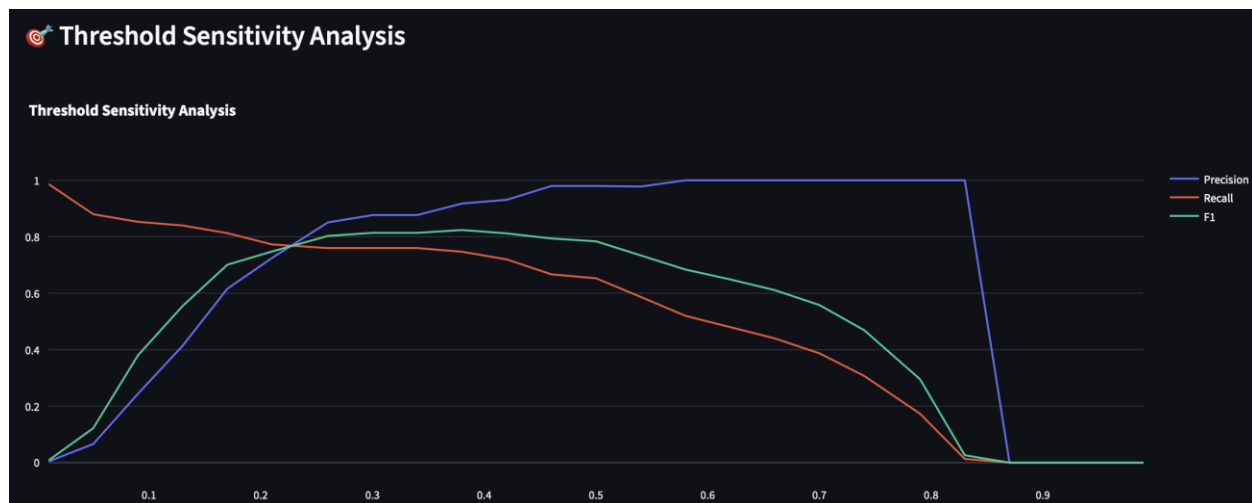


Figure 6.6 illustrates changes in predictive performance across varying classification thresholds.

## 6.6 Behavioural Drift Evaluation

Behavioural drift monitoring was evaluated to determine whether evolving transaction patterns could be identified effectively within operational environments. Drift monitoring generated indicators including drift scores, severity classifications, and feature-level behavioural changes to assess model stability over time.

Observed outputs indicated a drift score of **0.0149** with a **low severity classification**, suggesting relatively stable behavioural conditions during experimental analysis. Although severe drift behaviour was not detected, feature-level monitoring identified repeated changes within variables including **V10, V26, V12, V9, and V17**, indicating that certain transaction characteristics demonstrated greater sensitivity to behavioural variation. Experimental observations additionally indicated sensitivity within Page–Hinkley detection behaviour, where minor fluctuations occasionally generated multiple drift events. This behaviour suggests that drift thresholds require careful tuning because excessive sensitivity may introduce false drift alerts and increase unnecessary operational intervention.

The ability to identify behavioural changes provides important operational value because evolving transaction behaviour may reduce future prediction reliability. Early identification of drift patterns can therefore support proactive intervention before substantial degradation in predictive performance occurs.

Figure 6.7: Behavioural drift summary indicators.

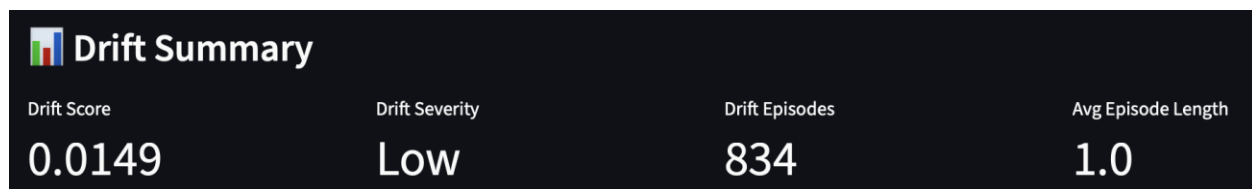


Figure 6.7 illustrates generated drift metrics including severity, score, and monitoring statistics.

Figure 6.8: Major behavioural drift drivers.

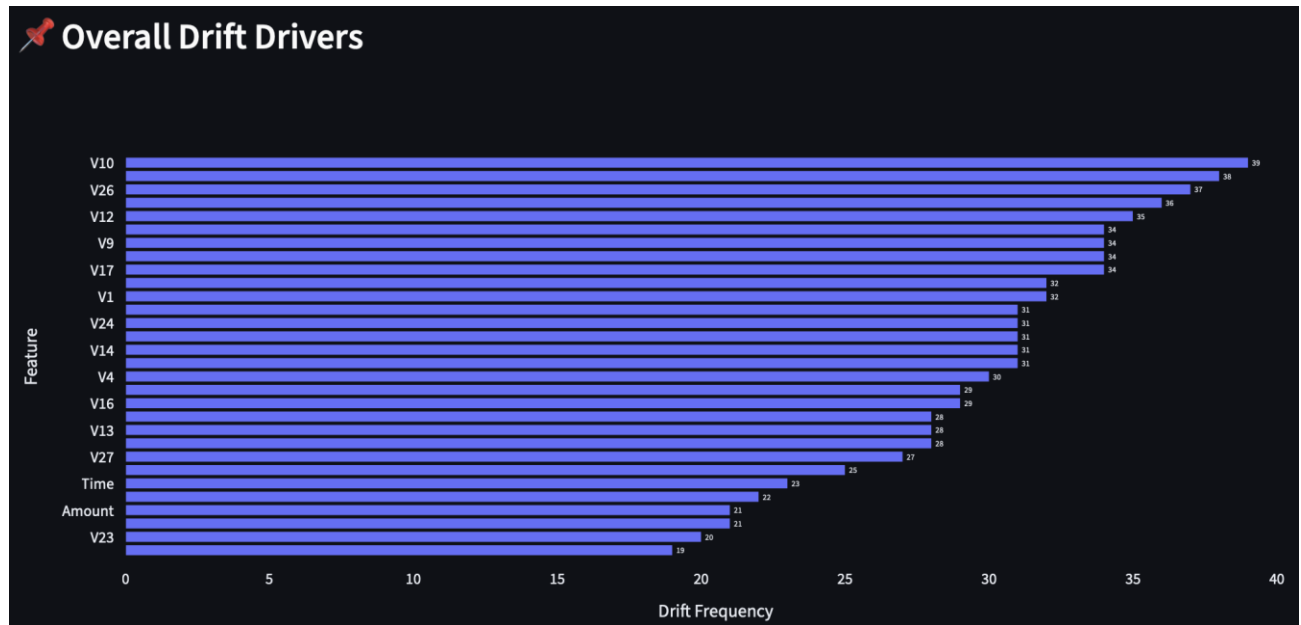


Figure 6.8 illustrates variables contributing most strongly toward detected behavioural variation.



## 6.7 Calibration Evaluation

Calibration evaluation was performed to assess whether prediction probabilities generated by the selected Random Forest model accurately reflected observed outcomes. Reliable probability estimates were considered important because governance routing decisions within FinSentinel depend partially on prediction confidence rather than binary classification outcomes alone.

Results indicated that the calibrated model achieved a **Brier Score of 0.0007**, suggesting strong agreement between predicted confidence values and actual transaction outcomes. Lower Brier scores generally indicate improved probability estimation quality, demonstrating that the model generated reliable confidence estimates across different prediction ranges.

The calibration curve further demonstrated that model predictions remained relatively close to the ideal calibration reference line. Minor deviations were observed at selected probability intervals; however, the overall trend indicates that predicted probabilities reasonably represented observed event frequencies.

These findings indicate that prediction confidence values remained relatively stable and aligned with observed outcomes. However, the current evaluation did not explicitly measure downstream effects on governance decisions. Therefore, calibration results should be interpreted as evidence of confidence consistency rather than direct operational improvement.

Figure 6.9: Model probability calibration analysis using calibration curves.

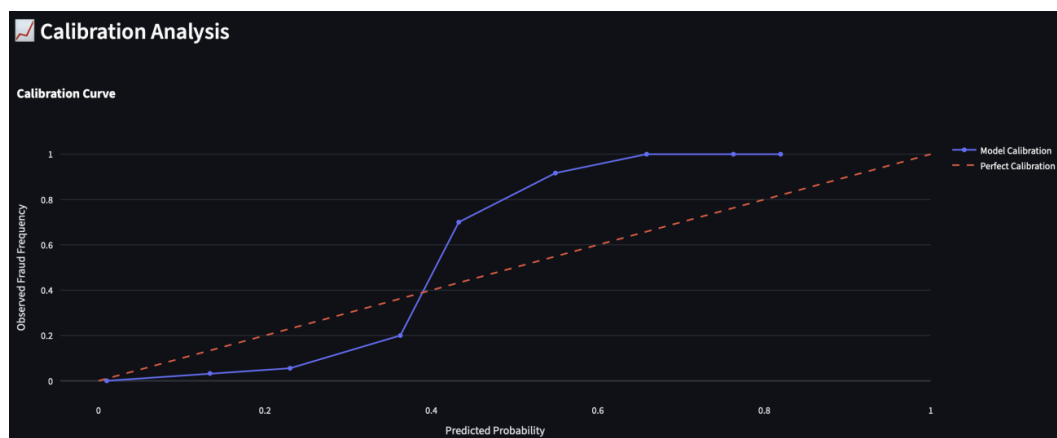


Figure 6.9 illustrates calibration performance and agreement between predicted probabilities and observed outcomes.

## 6.8 Human–AI Operational Evaluation

The Human–AI collaborative framework was evaluated to determine whether selective escalation mechanisms could improve operational performance within high-stakes fraud environments. Rather than replacing human judgement completely, uncertain transactions were routed toward analyst review while high-confidence predictions were processed automatically. The objective was to investigate whether collaboration between predictive intelligence and human oversight could improve fraud capture while reducing unnecessary intervention.

Results suggest that selective routing improved operational efficiency by reducing unnecessary analyst intervention while preserving fraud detection capability. However, increased automation may introduce dependence on model behaviour and therefore continued human oversight remains important.

The evaluation further demonstrated that combining predictive intelligence with governance mechanisms provided operational advantages beyond conventional standalone machine learning approaches. Human analysts were selectively integrated into the decision process rather than replacing automated functionality entirely.

These findings suggest that Human–AI collaboration within FinSentinel successfully balanced operational efficiency and human oversight by reducing unnecessary intervention while preserving control over uncertain transactions.

**Table 6.2: Human–AI Operational Impact Assessment**

| Operational Metric           | Observed Outcome |
|------------------------------|------------------|
| Auto Decision Rate           | 99.92%           |
| Human Review Rate            | 0.08%            |
| Fraud Intervention Rate      | 76.0%            |
| Wrongful Block Rate          | 0.0%             |
| Legitimate Intervention Rate | 0.01%            |

*Table 6.2 summarises operational performance indicators associated with Human–AI collaboration.*

## 6.9 Research Question Evaluation

Research questions were evaluated against findings obtained from predictive performance analysis, explainability mechanisms, governance experiments, and behavioural monitoring outputs. The objective was to determine whether the proposed framework successfully addressed the objectives defined during project development.

Evaluation results indicate that governance mechanisms improved operational control by incorporating decision support capabilities beyond prediction outputs alone. Results suggest that Human–AI collaboration may reduce analyst workload through selective escalation processes, while explainability mechanisms improved prediction transparency and interpretability. Behavioural monitoring additionally supported long-term system reliability through continuous observation of changing transaction patterns.

Although the framework successfully addressed the research objectives, findings indicate that operational effectiveness extends beyond predictive performance alone. Components such as governance policies and drift monitoring introduced additional trade-offs involving analyst workload, threshold sensitivity, and intervention behaviour. Consequently, practical deployment requires balancing technical performance with operational constraints.

**Table 6.3: Research Question Evaluation**

| Research Question                                                                                        | Evaluation Outcome                                                                                 |
|----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| RQ1: Can governance mechanisms improve trust and operational reliability in AI fraud systems?            | Governance controls improved operational decision support and confidence-based routing capability. |
| RQ2: Can Human–AI collaboration reduce analyst workload while maintaining fraud detection effectiveness? | Selective escalation reduced manual workload while preserving operational performance.             |
| RQ3: Can explainability mechanisms improve transparency of prediction outcomes?                          | SHAP explanations and analyst narratives improved interpretability of model decisions.             |
| RQ4: Can behavioural monitoring support long-term model reliability?                                     | Drift monitoring detected behavioural changes and supported proactive operational awareness.       |

*Table 6.3 summarises evaluation outcomes for the research questions investigated within the study.*

# Conclusion and Future work

## 7.1 Conclusion

The primary objective of this project was to develop and evaluate a governance-aware Human–AI collaborative fraud detection framework capable of improving transparency, operational control, and long-term reliability within fraud detection environments. Existing approaches frequently focus primarily on predictive performance while providing limited support for explainability, governance mechanisms, and behavioural monitoring. Consequently, the project aimed to address these limitations through the development of the FinSentinel framework.

The developed framework integrated multiple components including predictive intelligence, explainability functionality, governance routing mechanisms, behavioural drift monitoring, calibration analysis, and dashboard-based visualisation within a unified operational environment. Rather than functioning solely as a prediction system, FinSentinel was designed as a decision-support framework capable of combining machine intelligence with human oversight.

Evaluation results demonstrated that the selected Random Forest model achieved strong predictive capability, producing a **ROC–AUC value of 0.977**, **PR–AUC of 0.814**, and **F1-score of 0.814**. These findings indicate effective discrimination between legitimate and fraudulent transaction behaviour within highly imbalanced conditions.

The integration of SHAP explainability functionality improved transparency by identifying influential variables contributing toward prediction outcomes. Behavioural drift intelligence additionally provided awareness regarding changing transaction behaviour and supported long-term operational monitoring. Calibration mechanisms supported evaluation of confidence reliability by improving the relationship between prediction probabilities and observed outcomes.

Governance experimentation demonstrated that different threshold configurations produced varying operational consequences involving fraud capture capability, automation rates, and analyst workload. The results highlighted the importance of balancing predictive effectiveness with practical operational considerations.

Overall, the project successfully achieved its objectives and demonstrated that combining predictive intelligence with governance-aware Human–AI mechanisms can improve both technical and operational aspects of fraud detection systems.

## 7.2 Project Contributions

The project contributes to existing fraud detection research and implementation practice through several areas:

- Development of a Human–AI collaborative governance framework designed to investigate whether selective escalation of uncertain transactions can improve fraud capture while preserving operational efficiency.
- Integration of predictive intelligence, explainability mechanisms, and behavioural monitoring within a unified environment.
- Implementation of threshold-based governance routing for operational decision support.
- Incorporation of calibration mechanisms to improve confidence reliability.
- Development of an interactive dashboard supporting operational visualisation and analysis.

These contributions extend beyond predictive modelling by introducing additional intelligence mechanisms capable of supporting practical deployment environments.

## 7.3 Limitations

Although FinSentinel demonstrated promising performance, several limitations remain. The framework relied on a single publicly available dataset and therefore may not fully represent behavioural characteristics observed across real financial institutions. Since the dataset utilised anonymised PCA-transformed features, interpretation of feature behaviour becomes less meaningful compared with real-world transactional attributes.

Human analysts were simulated through governance logic rather than involving actual domain experts. Consequently, Human–AI collaboration outcomes may differ in practical operational environments where human judgement introduces contextual and subjective decision factors.

Additionally, behavioural monitoring relied primarily on simulated behavioural changes rather than continuous real-time transaction streams. Therefore, the proposed drift mechanisms should

be interpreted as demonstrating proof-of-concept capability rather than complete production readiness.

Finally, although Random Forest demonstrated strong predictive performance and interpretability, alternative adaptive learning systems may potentially demonstrate improved long-term performance under continuously evolving operational conditions.

Although transaction ordering was reconstructed using the dataset Time variable, the dataset did not represent true real-time streaming behaviour. Consequently, temporal behavioural patterns may differ from those observed within live financial systems.

The Page–Hinkley monitoring mechanism additionally demonstrated sensitivity to parameter selection. During experimentation, minor behavioural fluctuations occasionally generated multiple drift triggers, indicating that threshold selection may influence stability and potentially introduce false-positive drift alerts.

## 7.4 Future Work

Several opportunities exist for future enhancement of the proposed framework. Future work could involve evaluating the framework using additional financial datasets and real-world transaction environments to improve generalisability.

Real-time streaming architectures may additionally be integrated to support continuous monitoring of incoming transaction behaviour. Further research could investigate adaptive governance mechanisms capable of dynamically modifying threshold values according to changing behavioural conditions.

Additional Human–AI functionality including analyst feedback learning and reinforcement-based decision adaptation could also improve collaboration effectiveness. Future implementations may further incorporate Large Language Models and advanced explainability approaches to improve interaction and decision support.

## 7.5 Final Remarks

The increasing use of Artificial Intelligence within decision-critical environments requires systems that extend beyond predictive capability alone. Although high classification performance remains important, practical deployment additionally requires transparency, operational control, reliability, and human oversight. The FinSentinel framework demonstrated that combining predictive intelligence with governance-aware mechanisms can improve both technical and operational aspects of fraud detection systems.

The project therefore highlights the importance of Human–AI collaboration as a practical approach for supporting trustworthy Artificial Intelligence environments where predictive systems and human expertise operate together rather than independently.

# Reflection

## 8.1 Critical Reflection on Project Development

The development of FinSentinel provided practical experience across multiple areas including machine learning implementation, Human–AI collaboration, explainability mechanisms, governance design, and system integration. The project extended beyond developing predictive models and required consideration of broader operational and decision-support factors associated with practical deployment environments.

One significant learning outcome involved recognising that predictive accuracy alone may not represent sufficient evaluation for real-world fraud detection systems. During the early stages of development, emphasis was primarily placed on improving classification performance through model selection and comparative parameter experimentation. However, further investigation demonstrated that strong predictive performance does not necessarily ensure practical usability or trustworthy decision-making. Consequently, additional components including explainability functionality, governance mechanisms, and behavioural monitoring were incorporated into the framework.

Another important challenge involved managing severe class imbalance within the fraud dataset. Since fraudulent observations represented only a small proportion of total transactions, traditional evaluation measures occasionally produced misleading performance interpretations. This required the use of additional performance measures including precision, recall, F1-score, and PR-AUC to provide more reliable assessment.

The implementation process also highlighted challenges associated with integrating multiple independent components within a unified framework. Individual components such as predictive models, explainability modules, governance systems, and drift monitoring mechanisms functioned successfully in isolation; however, ensuring consistent interaction between modules required repeated testing and refinement. Iterative development therefore became necessary throughout implementation.

The incorporation of governance threshold experimentation additionally provided insight into the relationship between predictive outcomes and operational consequences. Different threshold configurations demonstrated that improving fraud detection capability could simultaneously increase analyst workload or introduce additional intervention risks. This highlighted the importance of balancing technical optimisation with practical organisational requirements.



Overall, the project provided valuable experience regarding both technical implementation and broader Human–AI system design principles. The development process reinforced the importance of transparency, operational trust, and continuous monitoring within Artificial Intelligence environments.

## 8.2 Lessons Learned and Personal Development

The project contributed to both technical and professional development. Practical experience was gained in machine learning implementation, explainability techniques, Human–AI interaction, model calibration, and behavioural monitoring.

The project additionally strengthened research, problem-solving, and project management skills. Iterative implementation required adaptation of design decisions according to emerging challenges, reinforcing the importance of flexibility and continuous evaluation in Human–AI systems.

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# Appendix

## Appendix A: Experimental Model Configurations

This appendix summarises the experimental model configurations used during comparative analysis within FinSentinel. Rather than performing automated hyperparameter optimisation procedures, multiple predefined configurations were manually constructed to investigate how parameter selection influenced predictive performance and operational governance outcomes. Model configurations were evaluated comparatively according to fraud detection performance and Human–AI operational behaviour.

### Appendix A.1 Logistic Regression Configurations

| Model       | Configuration                                          |
|-------------|--------------------------------------------------------|
| LogReg_C0.1 | C = 0.1, Max Iterations = 500, Class Weight = Balanced |
| LogReg_C1   | C = 1, Max Iterations = 500, Class Weight = Balanced   |
| LogReg_C10  | C = 10, Max Iterations = 1000, Class Weight = Balanced |

### Appendix A.2 Random Forest Configurations

| Model       | Configuration                                           |
|-------------|---------------------------------------------------------|
| RF_100_10   | Trees = 100, Depth = 10, Class Weight = Balanced        |
| RF_300_20   | Trees = 300, Depth = 20, Class Weight = Balanced        |
| RF_500_None | Trees = 500, Depth = Unlimited, Class Weight = Balanced |

### Appendix A.3 XGBoost Configurations

| Model       | Configuration                                                  |
|-------------|----------------------------------------------------------------|
| XGB_Default | Default configuration with Scale Positive Weight               |
| XGB_Tuned   | Trees = 300, Depth = 8, Learning Rate = 0.05, Subsample = 0.8  |
| XGB_Deep    | Trees = 500, Depth = 12, Learning Rate = 0.03, Subsample = 0.9 |

#### Appendix A.4 LightGBM Configurations

| Model         | Configuration                                 |
|---------------|-----------------------------------------------|
| LGBM_Balanced | Trees = 300, Learning Rate = 0.05, Depth = 8  |
| LGBM_Recall   | Trees = 500, Learning Rate = 0.03, Depth = 12 |
| LGBM_Deep     | Trees = 700, Learning Rate = 0.02, Depth = 15 |

#### Appendix A.5 CatBoost Configurations

| Model       | Configuration                                      |
|-------------|----------------------------------------------------|
| CAT_Default | Default Balanced Configuration                     |
| CAT_Tuned   | Iterations = 500, Learning Rate = 0.05, Depth = 8  |
| CAT_Deep    | Iterations = 700, Learning Rate = 0.03, Depth = 10 |

#### Appendix A.6 Multi-Layer Perceptron (MLP) Configurations

| Model       | Configuration                                                      |
|-------------|--------------------------------------------------------------------|
| MLP_Default | Hidden Layers = (100), Alpha = 0.0001, Max Iterations = 300        |
| MLP_Tuned   | Hidden Layers = (128,64), Alpha = 0.001, Max Iterations = 500      |
| MLP_Deep    | Hidden Layers = (256,128,64), Alpha = 0.0005, Max Iterations = 700 |

#### Appendix A.7 Experimental Evaluation Criteria

The experimental configurations were evaluated using predictive and operational performance indicators including:

- ROC–AUC
- PR–AUC
- Precision
- Recall
- F1 Score
- Brier Score
- Threshold Sensitivity Analysis
- Fraud Capture Rate
- Human Review Rate
- Governance Performance Metrics

